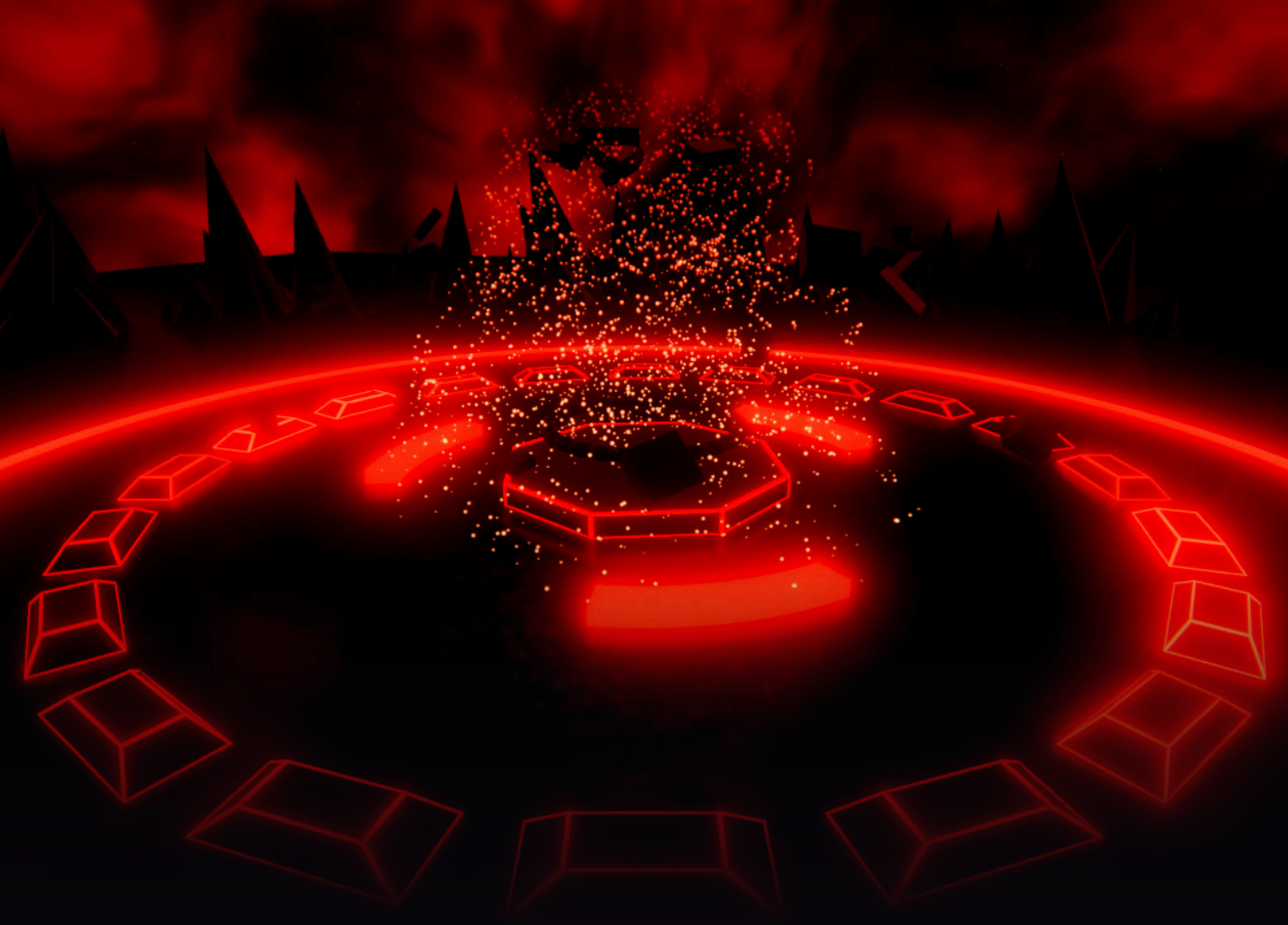


USER MANUAL · VERSION 0.6

EZ2DEMOSCENE

Build seamless, stylised 3D loops the demoscene way: neon, mirrors, particles and retro pixels. No code needed.

For Windows, Linux, macOS, the web & Android · Covers the editor, the node graph, exporting and the command line

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11 Randomize, Surprise me & undo

12 Nodes mode (advanced)

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15 Phones, tablets & the web

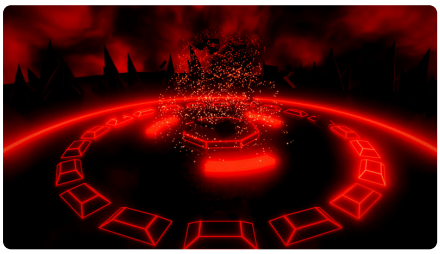
16 Recipes

17 Reference: shortcuts, performance, troubleshooting, glossary

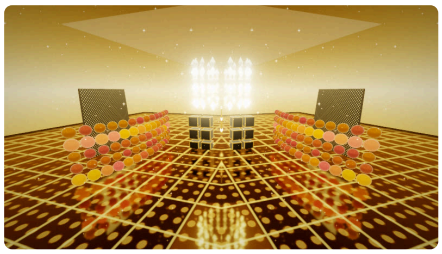
1 Welcome

EZ2DEMOSCENE is a tool for making **short, perfectly looping 3D animations** in the style of the demoscene: glowing neon arenas, mirrored golden halls, glossy crystals orbited by debris, raymarched tunnels, kaleidoscopes and synthwave sunsets. The loops are for music visuals, VJ sets, wallpapers, social posts and GIFs.

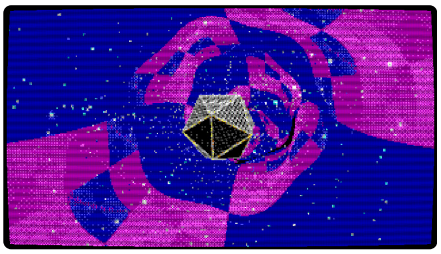
You never write code or shaders. You pick a starting scene, then change it with sliders, colour pickers and menus. When you like it, you export an MP4, WebM, GIF or a folder of PNG frames.



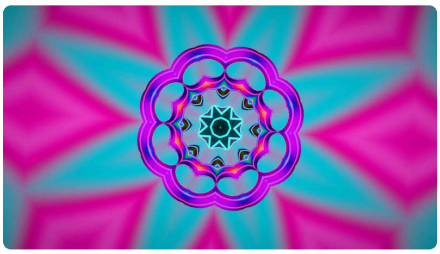
Neon Arena
Black gloss, red neon, crystal skyline.



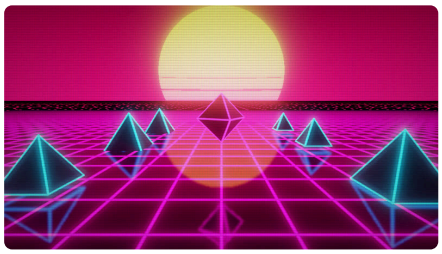
Gold Kaleido Room
Mirrored hall, orb walls, chandelier.



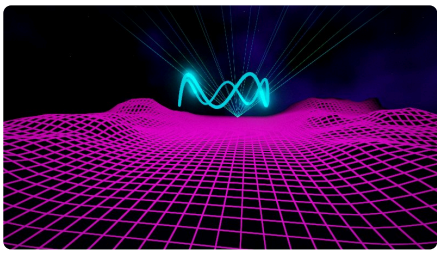
Retro Tunnel
XOR tunnel, EGA palette, CRT.



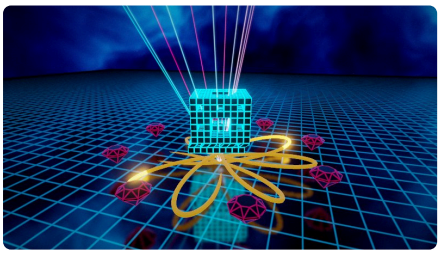
Plasma Kaleidoscope
Oldschool plasma, spinning tori.



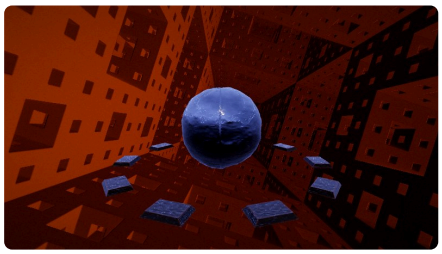
Synth Sunset
Striped sun, neon grid, pyramids.



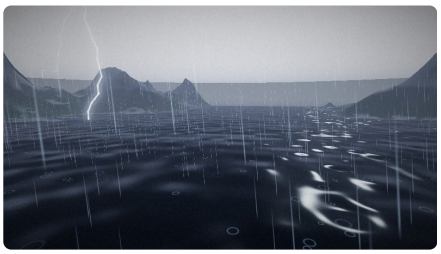
Vector Valley
Wireframe landscape, laser fans, neon ribbon.



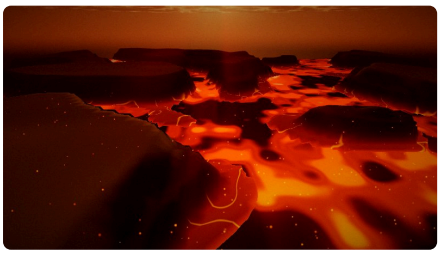
Glitch Shrine
Glitching sponge, flashing gems, pulsing rose.



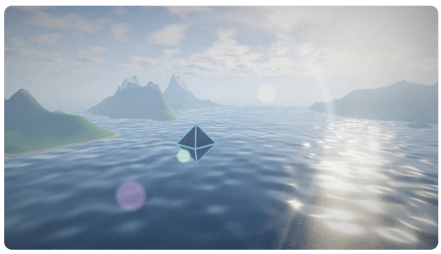
Sponge Dive
Menger sponge flight, breathing displaced orb.



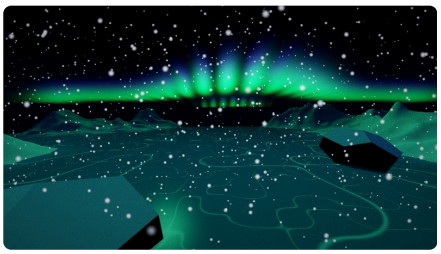
Stormy Lake
Rain, lightning and storm clouds over a lake.



Lava World
Lava rivers in volcanic canyons, embers.



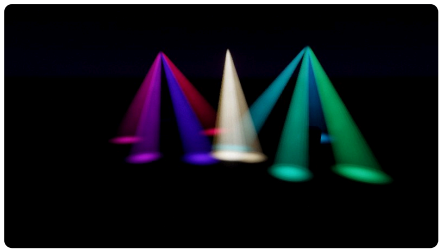
Sunbeam Peaks
Volumetric clouds, god rays, lens flare.



Aurora Tundra
Northern lights, frozen lake, snow.



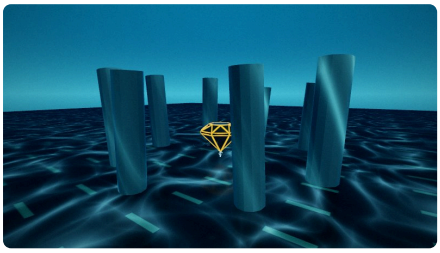
Dune Sea
Sand dunes, sandstorm, a toxic oasis.



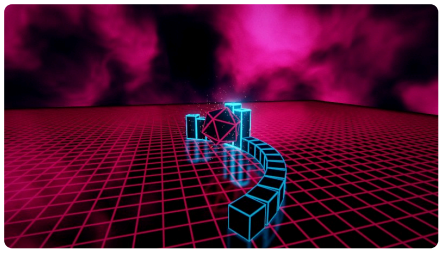
Club Spotlights
Spotlight cones, light pools, mist.



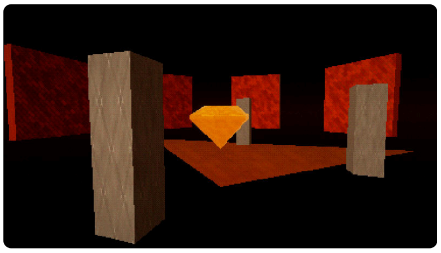
Rainbow Falls
Day and night, a rainbow, a waterfall.



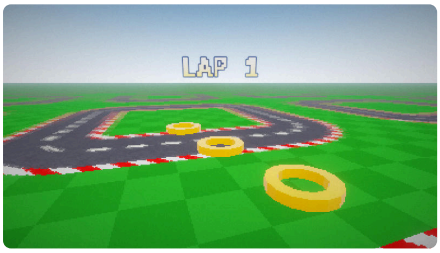
Sunken Temple
Underwater caustics and bubbles.



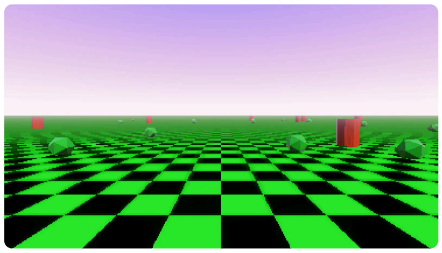
Music Reactor
Load a song: equalizer, kick flashes, time warp.



PSX Crypt
PlayStation: wobbling polygons, swimming textures.



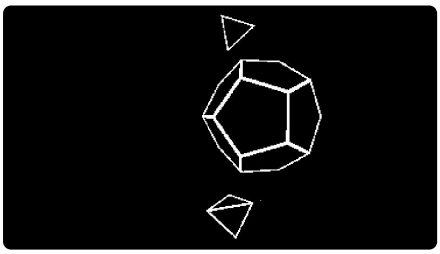
Mode 7 Circuit
SNES racing on a turning, endless track.



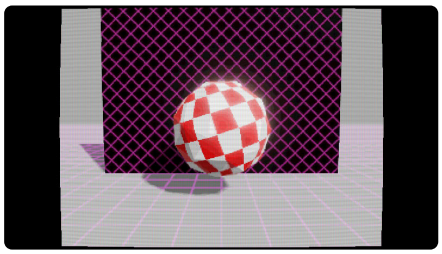
Harrier Plains
A chequered floor rushing to the horizon.



Slipgate Courtyard
Quake: palette lighting, torches, lava.



Wireframe Trader
1-bit wireframe ships and a space station.



Amiga Bounce
The red and white ball, bouncing on a grid.



C64 Intro
Raster-barred logo and a sine scroller, 160 × 200.

The one big idea: everything loops

A scene is exactly **a whole number of beats** long, at a tempo you choose (BPM). Every motion in EZ2DEMOSCENE completes a whole number of times during that loop. That includes spins, orbits, pulses, particles, scrolling textures, drifting clouds and even film grain. So the last frame leads straight back into the first, with no jump.

> **Why this matters**
>
> You never have to "fix the loop" by hand. Whatever you change, the export repeats seamlessly. This is why many settings ask for a count of *turns per loop* or *cycles per loop* rather than a speed.

How a scene is built

A scene is a stack of **layers** drawn together:

- **Background**: the sky or environment behind everything (nebula, tunnel, plasma…).
- **Mirror floor**: a reflective ground plane, optionally with a glowing grid.
- **Shapes**: 3D objects, possibly copied many times in rings, grids, walls or swarms.
- **Particles**: sparks, stars, glitter, vortices.

On top of the layers sit the **camera**, the **light & fog**, and the **post effects** that give the final look (glow, kaleidoscope, retro palettes, CRT…).

2 Installing & launching

> **No install needed**
> You can also use EZ2DEMOSCENE in a web browser or on an Android phone. See chapter 15.

What you need

Item	Details
A graphics card	Any GPU with Vulkan, DirectX 12 or Metal support (most GPUs from 2015 onward). A software renderer also works, only slower.
Rust 1.95 or newer *(source builds only)*	Needed to build the app from source. Install it from `rustup.rs`, or update with `rustup update stable`.
ffmpeg *(optional)*	Needed only for MP4, WebM and GIF export. Download it from `ffmpeg.org` and make sure the `ffmpeg` command works in a terminal. PNG sequences need nothing extra.
Linux only	`sudo apt install libasound2-dev libxkbcommon-dev libwayland-dev`

The easy way: download a release

Ready-made downloads are published on the project's GitHub **Releases** page for Windows, Linux and macOS. Unzip the archive and run `ez2demoscene` (`ez2demoscene.exe` on Windows). The Windows and Linux downloads include **ffmpeg**, so video and GIF export work straight away. On macOS, install it with `brew install ffmpeg`. The archive also contains this manual, the presets and the texture pack.

Building from source

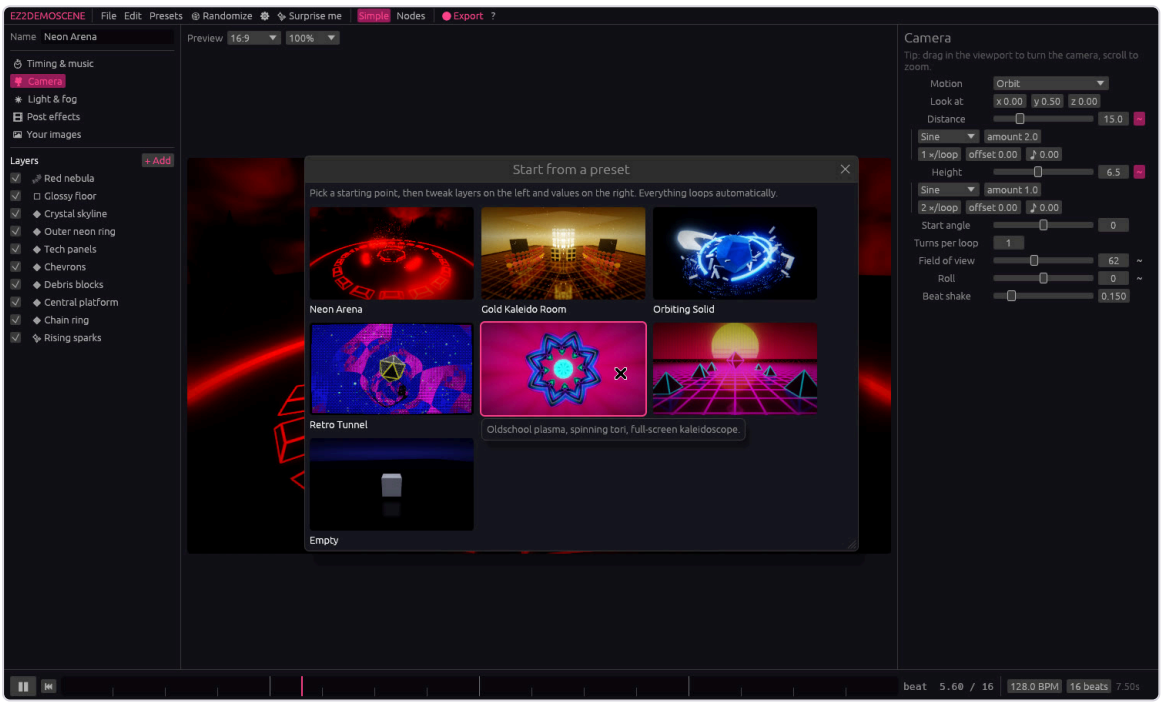
```
git clone https://github.com/Roganis/EZ2DEMOSCENE
cd EZ2DEMOSCENE
cargo run --release                  # opens the editor
cargo run --release -- my.ez2.json   # opens a saved project
```

The first build downloads and compiles libraries, so it takes a few minutes. Later starts are instant. The finished program is `target/release/ez2demoscene` (`ez2demoscene.exe` on Windows), and you can copy it anywhere.

The editor opens on an empty stage, paused (press **Space** or ▶ to play), with the **preset gallery** on top. Choosing a preset loads it and starts playing. Tick **Don't show on startup** at the bottom of the gallery to skip it; it stays one click away under **Presets** in the menu bar. The built-in presets come in five groups:

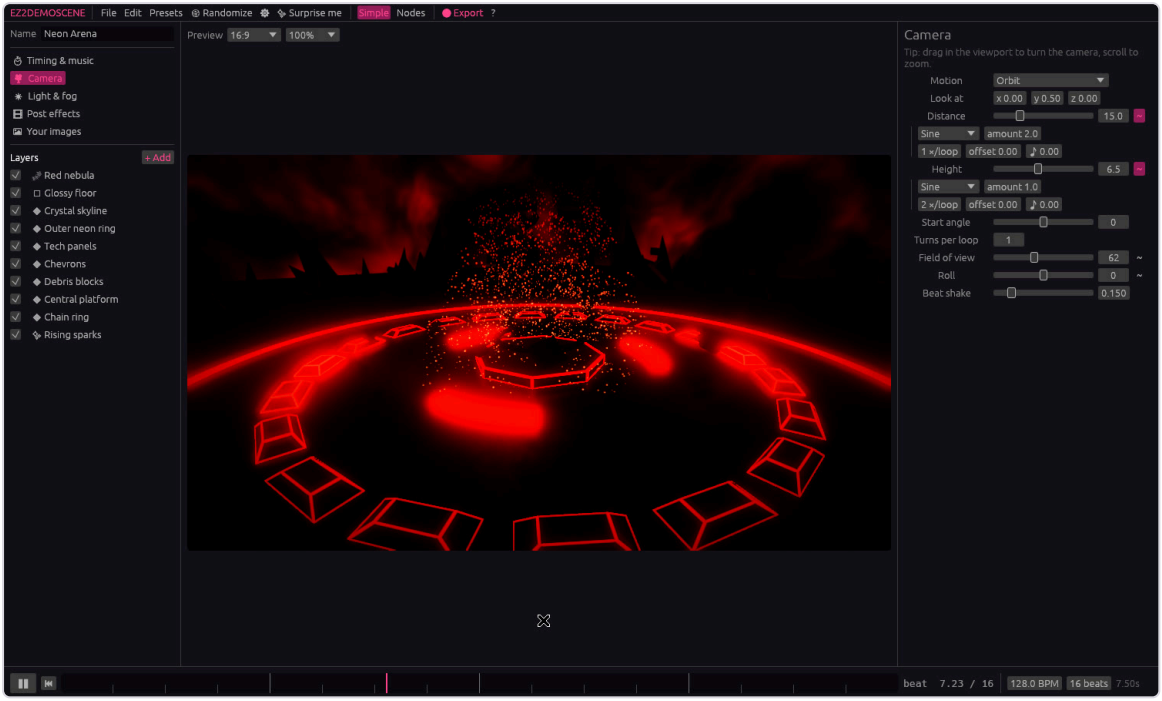
- **Consoles & arcade**: PSX Crypt (PlayStation), Saturn Ghosts, Fog Island (Nintendo 64), Mode 7 Circuit (SNES), Super FX Starship (SNES Star Fox), Harrier Plains (Space Harrier arcade), Polygon Arcade (Sega Model 1), Battle Screen (16-bit RPG), Red Visor (Virtual Boy) and Dreamcast Sunset.

- **PC era**: Slipgate Courtyard and Slime Falls (Quake), Hangar Base (early-90s software 3D), Accelerator Arena (late-90s 3D cards), Wireframe Trader (BBC Micro Elite) and Screen Saver.
- **Home computers & handhelds**: Amiga Bounce, C64 Intro, Spectrum Isometric, Tape Loader (a Spectrum loading from tape) and Handheld Quest (Game Boy), each at its machine's resolution, in its palette and inside its border.
- **Demoscene**: the classic effects: tunnels, plasmas and kaleidoscopes, Copper Heaven (an Amiga demo), Demo Party 93 (a 1993 PC demo on a timeline), Rotozoomer, Vector Balls, Glenz Vectors and Oldschool Fire.
- **Engine showcases**: one idea each: physical materials, reflections and light shafts, simulated liquids, cloth, rigid bodies and flocks, weather and skies, music, nodes, the timeline and stepped animation.



The preset gallery appears at start-up. Hover a card for a description, click to load it.

3 A tour of the window



The editor with the Neon Arena preset and the Camera selected.

Area	What it does
Top bar	**File**, **Edit**, **Presets**, 🎲 **Randomize** (with its ⚙ settings), **Surprise me**, the **Simple** / **Nodes** mode switch, ● **Export** and **?** help.
Scene panel (left)	The project **name**; the scene-wide sections (**Timing & music**, **Camera**, **Light & fog**, **Post effects**, **Retro 3D**, **Your images**); and the **layer list** with **+ Add**.
Viewport (centre)	A live preview. **Drag** to turn the camera and **scroll** to zoom. Above it you can pick the **aspect ratio** (16:9, 4:3, 1:1, 9:16, 21:9), the **preview quality**, a live **fps** readout, the **Move** / **Rotate** / **Scale** tools and the **Grid** switch (see 5.6). Click an object to select it. Status messages also appear here.
Inspector (right)	All the settings of whatever is selected on the left.
Timeline (bottom)	Play/pause, back to start, a scrubber with **beat ticks** (tall ticks mark every 4 beats), the current beat, BPM and loop length.

> **Tip**
> Hover almost any label in the inspector to see a short explanation.

4 Quick start: your first loop in five minutes

1. **Pick a preset.** In the gallery, click **Tesla Swarm**. The scene starts playing straight away.
2. **Change a colour.** Click the **Dodecahedron** layer on the left. In the inspector under **Material**, click the **Colour** swatch and pick a hot pink.
3. **Make it glow on the beat.** Still in **Material**, find **Glow** and click the **~** button next to it. Set the wave to **Pulse**, **amount** to 3 and **×/loop** to **8** (the loop is 8 beats long). It now flashes on every beat.
4. **Add particles.** Click **+ Add → ✨ Particles → Vortex**. In the new layer, set **Colour (young)** to white and **Colour (old)** to pink.
5. **Add a retro touch.** Select **Post effects** on the left and tick **Retro palette**. Choose **PICO-8**.
6. **Export.** Press `Ctrl`+`E`. Pick **Animated GIF** (the size switches to 480×270, a good size for GIFs), choose a file with **Browse…** and press **● Export**.
7. **Save your project** with `Ctrl`+`S` to keep editing it later.

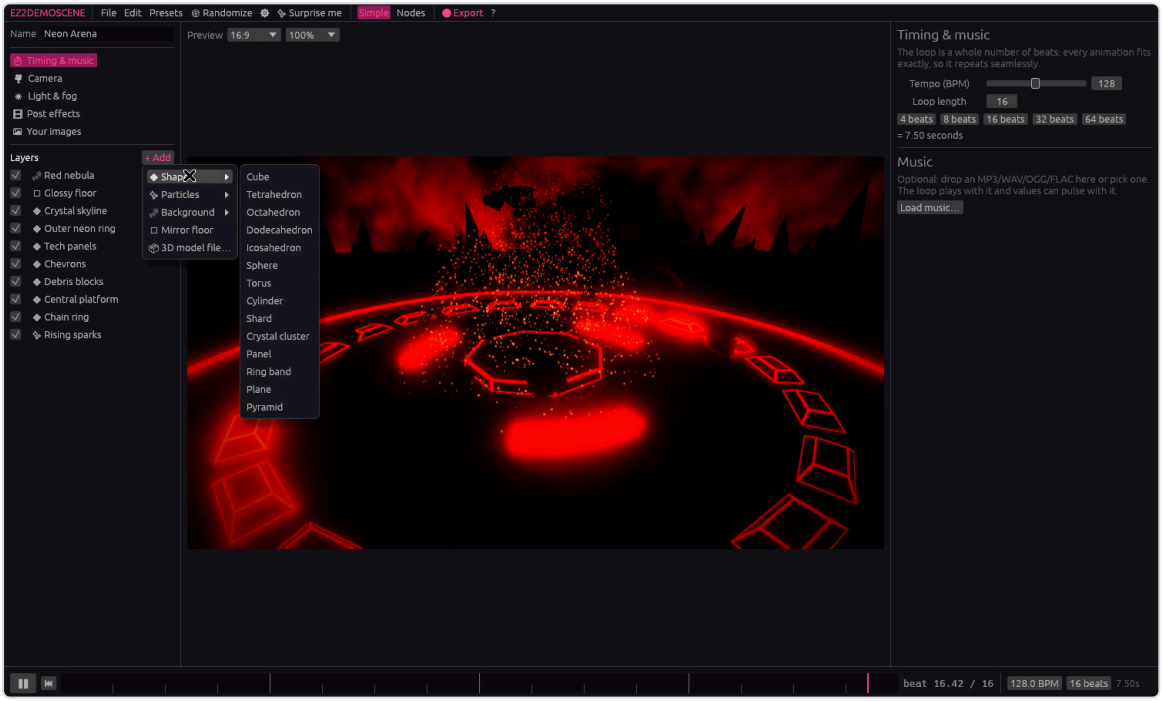
> **Made a mess?**
>
> `Ctrl`+`Z` undoes any change, including randomizing and loading presets.

5 Layers

The layer list works like the layers in a paint program. Each row has a **visibility** tick box and the layer's name. Click a row to edit it. Under the list, the buttons ⬆ ⬇ ⧉ 🗑 move, duplicate or delete the selected layer, and **right-clicking** a layer shows the same options.

Keys (with a layer selected)	Does
`Ctrl`+`C` / `Ctrl`+`X`	Copy / cut the layer. It goes on the clipboard, so you can also paste it into another project.
`Ctrl`+`V`	Paste a copied layer just after the selected one (at the end if none is selected).
`Ctrl`+`D`	Duplicate the layer.
`Delete` (or `Backspace`)	Delete the layer. `Ctrl`+`Z` brings it back.

They work in the **Simple** view while you aren't typing in a field. On a Mac, use `Cmd` for `Ctrl`.



+ Add offers shapes, particles, backgrounds, terrain, lasers, ribbons, weather, a mirror floor and 3D model files.

5.1 Backgrounds

A background fills the whole screen behind the other layers and follows the camera. Every background has three colours (**Colour A** is usually the darkest), a **Motion / loop** count, a **Brightness** and a **Detail** (pattern scale).

Style	Description
Gradient sky	A soft two-colour sky with a horizon glow and a sun halo in the light direction.

Nebula clouds	Swirling fractal clouds with stars. The clouds drift on a closed path so they loop.
Starfield	Twinkling stars over faint coloured dust.
Raymarched tunnel	The classic demo tunnel flying forward through a winding tube. Pick a **Wall texture** (the XOR or checker textures are great).
Raymarched fractal	A volumetric "Kaliset" fractal space. It looks psychedelic and cosmic, and is heavy on slow GPUs.
Oldschool plasma	The 90s sine plasma.
Synthwave sun & grid	A striped sun sinking into a scrolling neon grid, with stars.
Raymarched sponge flight	Flying through an endless Menger sponge, a lattice of beams or a field of cubes. Heavy on phones: lower **Quality** if it stutters.
Raymarched ring corridor	Flying through glowing rings, squares or triangles that alternate between colours B and C.
Volumetric clouds	A real sky: a gradient from colour B (horizon) to A (overhead), a sun in the light direction and raymarched 3D clouds lit by it, glowing at the edges when you look towards the sun. Colour C tints the shadowed sides. The clouds drift with **Motion / loop**. Pair it with **God rays** for sunbeams. Heavy on phones.
Aurora night sky	Stars and rippling northern-lights curtains that fade from colour B at the bottom to colour C at the top. Colour A is the night sky.
Battle background (retro RPG)	Flat on the screen like the battles of 16-bit role-playing games: tiled patterns whose lines wobble while the colours A → B → C cycle through them. See below.
Environment map	The map of the environment light (**Light & fog → Environment light**): the studio or panorama photo that lights the shapes, turned with it. **Sharpness** blurs it (lower it to keep the eye on the foreground). With the environment light on Colours it draws the sky and ground colours.
Two-layer sky (Quake)	Quake's sky: a far and a near picture on a flattened dome, stretching towards the horizon. The near layer is drawn over the far one except where it has the **See-through colour** (black by default, within a **Tolerance**). Each layer scrolls a whole number of tiles per loop (**Far scroll / loop**, **Near scroll / loop**: the near one usually faster). **Tiles** sets the pattern size and **Flatten** how flat the dome is (Quake used 3). Leave the layers on None for the built-in `sky_far` and `sky_near` clouds, or pick any texture or your own images (Quake skies were one picture split in two halves: save each half).

> **Tip**
> You can stack two backgrounds, but the later one covers the earlier one (and only that one is drawn). Usually one is enough.

Resolution renders a background at half or quarter resolution and upscales it. For volumetric clouds and the raymarched styles this is 4× or 16× less work and hardly visible; new clouds, fractal

and sponge backgrounds start at half.

Raymarching settings

The four raymarched backgrounds (tunnel, fractal, sponge, rings) get an extra **Raymarching** section. Only the settings that matter for the chosen style are shown, and all of them except Variant, Wall pattern, Roll and Quality can be animated with **~**.

Setting	Tunnel	Fractal	Sponge	Rings
Variant	Round, Square, Hexagon, Triangle, Flower	Kaliset, Crystal, Nebula	Menger sponge, Beam lattice, Cube field	Rings, Squares, Triangles
Size	Radius	Zoom	Cell size	Ring size
Twist	Twists the walls along the flight	n/a	Twists the space	Turns each ring a little more
Warp	Wall wobble	Fold (changes the whole look)	n/a	Thickness
Bend	How much the path winds	Drift	Path sway	Path sway
Glow	Light rings sliding past	Brightness	Edge glow	Glow
Fog ×	Depth fog (0 = none).			
Roll / loop	Whole turns of the view per loop.			
Quality	Raymarch steps (fractal: iterations). 0 uses the default; lower it for phones.			

The tunnel also has a **Wall pattern** (XOR, Checker, Rings, Stripes) used when there is no wall texture.

Sky settings

The two skies share the same section, with their own meanings:

Setting	Volumetric clouds	Aurora night sky
Variant	Fluffy, Overcast, Storm (dark and heavy)	Curtains, Rings (around the zenith), Spiral
Size	Cloud scale	Height of the curtains
Twist	n/a	Sway: how much the curtains swing over the loop
Warp	Coverage: how much of the sky is cloudy	Ripples: the fine vertical rays
Bend	Thickness of the cloud layer	n/a
Glow	Sun glow around the sun and through the clouds	Brightness

Fog ×	Haze: how much distant clouds melt into the horizon	n/a
Quality	Raymarch steps (default 28)	n/a

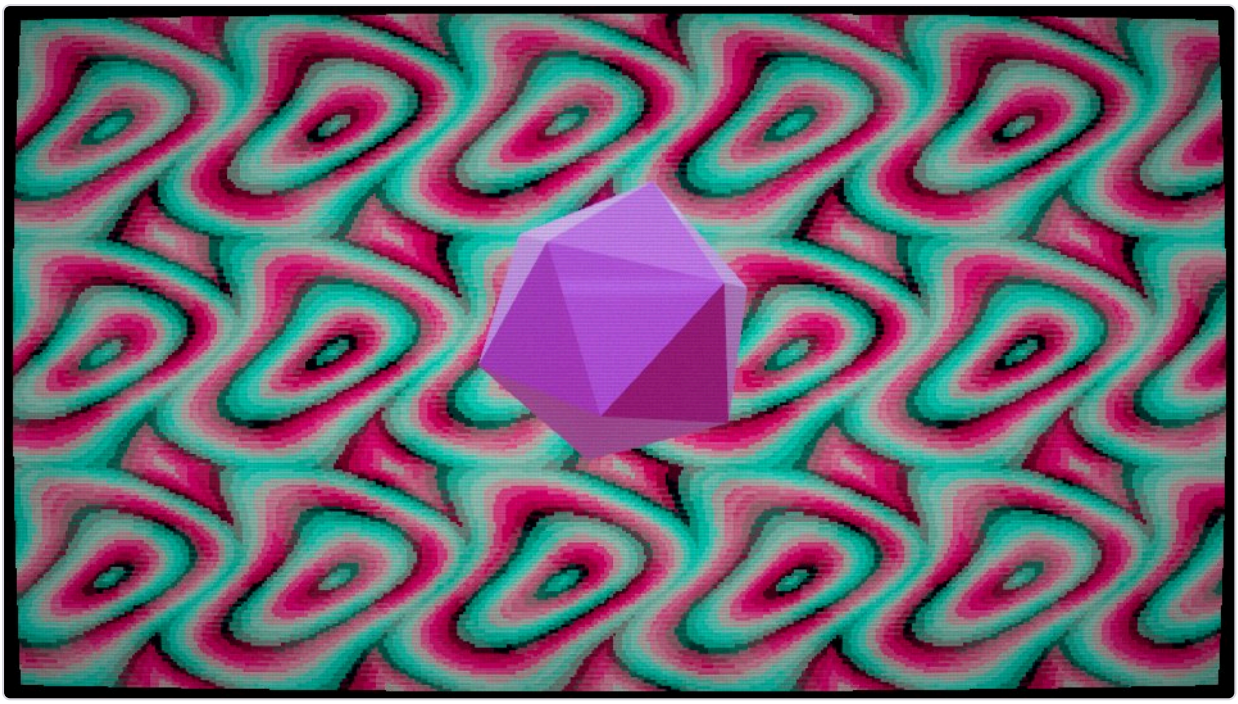
> **Loops**
>
> Clouds can drift any distance and still loop: two copies of the cloud field, half a loop apart, cross-fade into each other, so the jump back at the end of the loop happens while it is invisible.

Battle background

The **Battle background** style is a flat, retro picture behind everything, made of a **Back layer** and an optional **Front layer** over it. Each layer is a tiled **Pattern** (rings, diamonds, checker, diagonal stripes, zigzag, dots, swirl, bricks, plasma, or your own **picture**, whose brightness picks the colour). The colours A → B → C → A run round along the pattern and cycle through it, the classic colour-cycling animation.

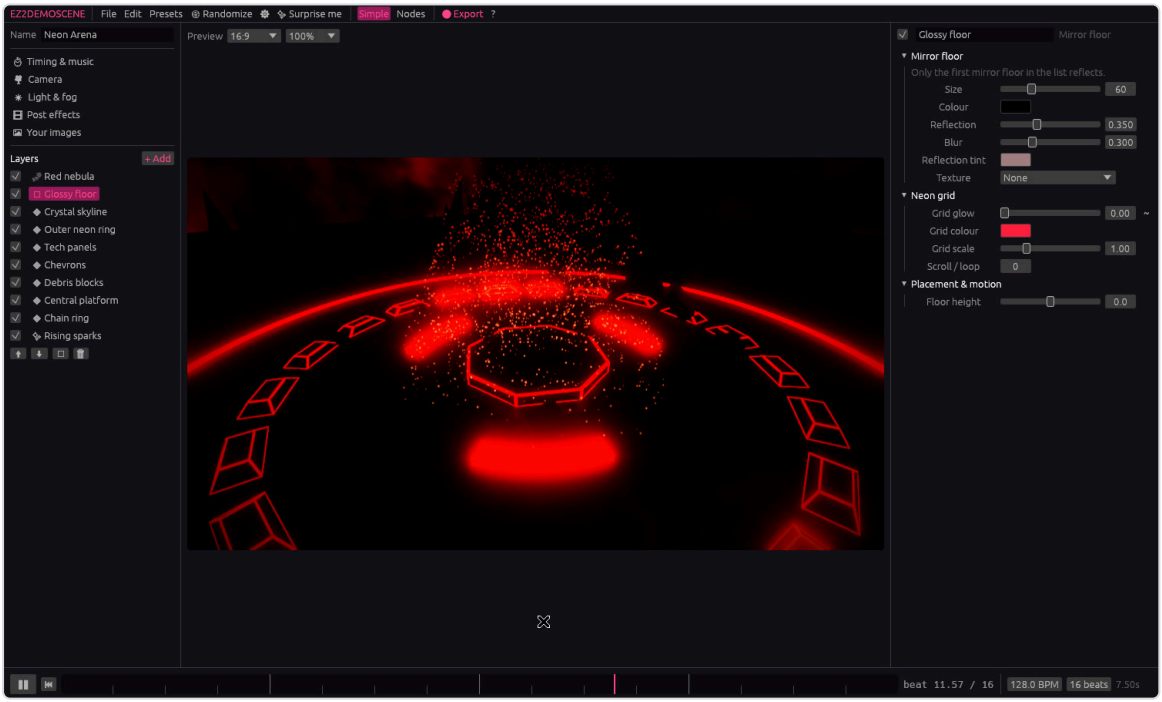
Setting	What it does
Lines	Pixel rows from top to bottom: fat retro pixels (224 is a Super Nintendo screen; 0 = full resolution).
Colours	How many colours the cycling palette has: a few give hard retro bands, 0 is smooth.
Tiles	How many times the pattern repeats from top to bottom.
Scroll / loop	Whole tiles the pattern scrolls per loop, sideways (x) and up (y).
Line warp	**Wave**: each line slides left and right along a wave. **Interlaced**: odd and even lines slide opposite ways, the shimmering battle look. **Compression**: lines bunch up and spread out vertically.
Amount, Waves, Waves / loop	How far lines move, how many waves fit from top to bottom, and how many times they roll by per loop.
Bands	How many times the colours go round across one tile.
Colour cycles	How many times the colours cycle through the pattern per loop (negative: the other way).
Blend, Opacity	Front layer only: **Mix**, **Add**, **Screen** or **Difference** over the back layer, and how much of it shows.

Everything that moves turns a whole number of times per loop, so the loop is seamless, and the numbers have ~ and 🎵 as usual. The **Battle Screen** preset puts a spinning crystal foe in front of one.



Battle Screen: rings cycling through three colours with a wave warp, a second pattern added over them, on a CRT.

5.2 Mirror floor



Mirror floor settings.

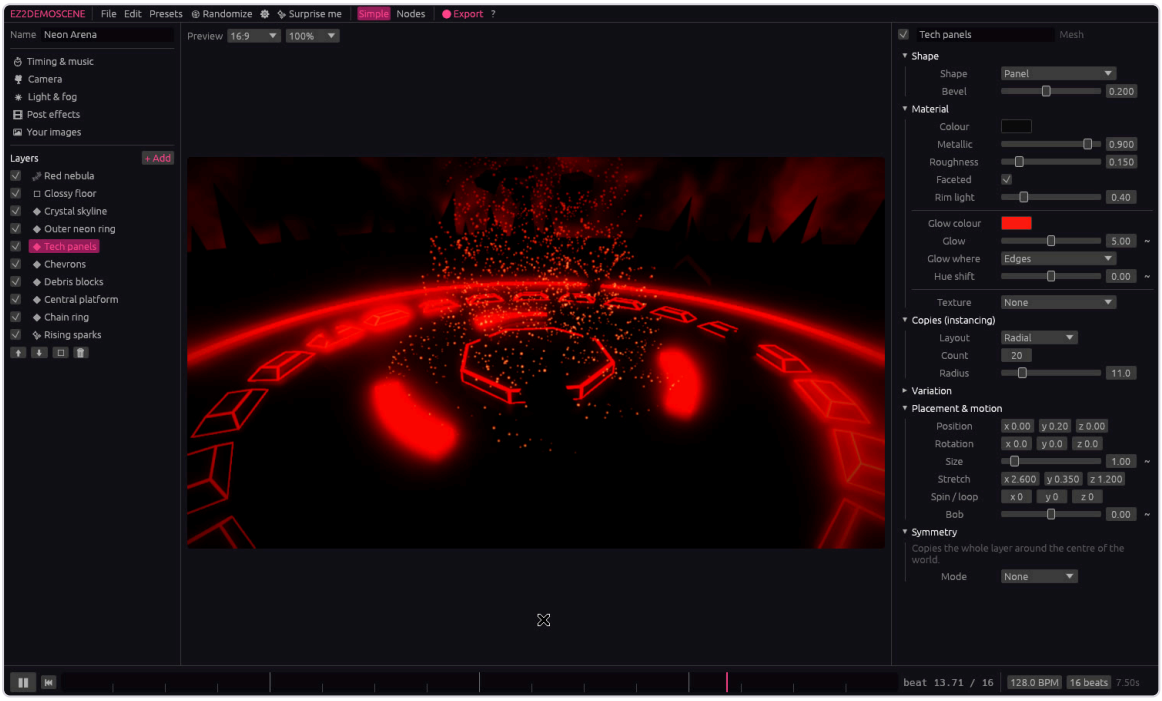
A large reflective ground plane. The scene is really reflected in it, including particles and the sky.

- **Infinite** stretches the floor to the horizon: it follows the camera and fades into the sky there, so no edge or corner ever shows. **Size** sets a square floor instead.
- **Reflection**: 0 is matte, 1 is a perfect mirror. **Blur** gives a frosted, polished-stone look. **Reflection tint** colours the reflection (gold for a gold room).
- **Texture** and **Tiling** put a pattern on the floor. The `led_grid` texture makes an LED dance floor.

- **Neon grid** draws glowing lines. Set **Grid glow**, **Grid colour**, **Grid scale** and **Scroll / loop** (cells per loop) for the synthwave "driving" effect.
- **Floor height** (in **Placement & motion**) moves the floor up or down.

> **One mirror**
>
> Only the **first** mirror floor in the list reflects. Extra ones are ignored.

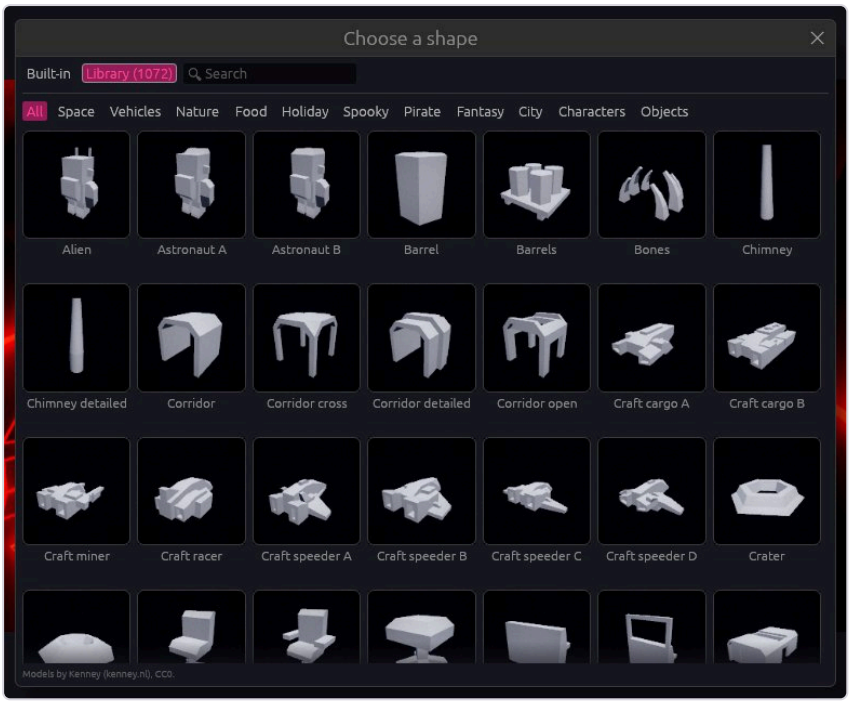
5.3 Shapes



A shape layer (Tech panels): Shape, Material, Copies, Variation, Placement & motion, Symmetry.

Click the **Shape** button to open **Choose a shape**, a window of tiles with a picture of every shape. The **Built-in** tab has the 25 built-in shapes, the raymarched ones, 3D text, cloth and **3D model file…**; the **Library** tab has about 1,000 ready-made low-poly models (spaceships, vehicles, nature, food, buildings, characters and more), with categories and a search box.

> **Library models**
>
> The models are by Kenney (kenney.nl) and free to use for anything (CC0). They take the layer's material, like every shape. The desktop apps carry the library; the web app downloads it (about 6 MB) the first time you open the Library tab.



The Library tab of **Choose a shape**: categories, search and a preview of every model.

The built-in shapes:

Shape	Good for
Cube, Tetrahedron, Octahedron, Dodecahedron, Icosahedron	Classic demo solids. Try them with **Faceted** on and **Glow where: Edges**.
Sphere (Detail 0–5)	Orbs and bubbles. Detail 0 gives a faceted low-poly ball.
Torus	Rings and chain links (Thickness, Segments).
Cylinder (Sides)	Platforms and pillars. Eight sides makes an octagonal stage.
Shard, Crystal cluster	Sharp spikes and crystal mountains. **Seed** gives a different random shape.
Panel (Bevel)	Sci-fi plates, light panels, screens.
Ring band (Arc, Width, Height)	Neon rings and arcs. Set Arc below 360° for partial arcs and chevrons.
Plane, Pyramid, Cone	Simple building blocks.
Capsule (Length), Rounded cube (Roundness)	Soft, toy-like objects and pills.
Torus knot (Loops, Twists), Spring (Turns), Möbius strip	Twisty tubes and bands. Knots with more loops and twists get more intricate.
Bowl (Thickness)	The lower half of a sphere as an open shell: a container for a Liquid layer.
Star (Points, Inner radius), Gear (Teeth), Heart	Flat extruded logos and badges (**Depth** sets the thickness).

Gem (Facets)	A brilliant-cut diamond. Lovely with **Glow where: Edges**.
Menger sponge (Level 0–3)	The classic fractal cube. Level 3 has 8,000 cubes, so use it sparingly.
3D text	Solid letters for logos: type the text (several lines are fine), pick a font (Pixel gives chunky voxel letters) or a TTF/OTF file, and set the **Depth**. They take every material, relief, glitch, deform and copies option.
Raymarched shapes	**Metaballs** (balls melting together as they orbit), **Gyroid** (a ball carved into a flowing lattice), **Fractal bulb** (the Mandelbulb) and **Melting box** (two balls passing through a rounded box). They are worked out per pixel, so they are perfectly smooth, cut into other shapes, floors and terrain correctly, cast and take sun shadows and use the layer's colour, metal, roughness, rim, glow and colour ramp (with **Glow where** other than Whole, the creases glow). **Motion** sets how many times their own movement repeats per loop (0 = still), and copies each get their own variation. Textures, relief, deform and glitch don't apply. They cost per pixel they cover, so a few big ones or a dozen small ones are fine; the **Liquid Metal** preset shows both.
Cloth	A sheet of cloth moved by the wind: a **Flag** (held by a pole along one side), **Curtain** (held along the top), **Banner** (held by its two top corners) or a **Drape over a ball** (a tablecloth or a ghost's sheet, gripping the ball, with an optional floor). Set its **Size**, **Detail** (finer folds, slower), **Stiffness** (silk to canvas) and the **Wind**: speed and direction are animatable and music-linkable (an oscillator on **Direction** swings it round; a pulse on the speed gusts on the beat), and **Gusts** make it vary over the cloth. The motion is **simulated** ahead of time into a loop, like flocks (*Simulating...* in the viewport bar; until then the cloth hangs at rest, and exports wait). The picture on the cloth is the material's texture, stretched once over the sheet; every material, copies, shadows and depth of field apply, so a Grid of one flag makes a row of flags. The **Banners** preset has five.
3D model file	Your own `.gltf`, `.glb`, `.obj`, `.stl`, `.ply`, `.off` or `.3mf`. It is centred and scaled to fit automatically. STL and 3MF files (3D printing) are Z up and are stood up the way the others are; files without normals get smooth ones that stay sharp across edges (a box stays a box). A glTF model brings its own material: colour, metal, roughness, glow and glass, and its pictures (colour, occlusion/roughness/metal, glow) are added to **Your images**, with **Physical** shading. Change any of it afterwards. OBJ models use the layer's material.

Copies (instancing)

One layer can show many copies of its shape. Pick a **Layout**. On desktop and in WebGPU browsers the graphics card places the copies, variation included, so even big counts cost the processor almost nothing; in WebGL2 browsers the processor does the same maths and gets the same picture, just slower. (Copies on a shape's surface or on a terrain are placed by the processor everywhere, then finished on the graphics card.)

Layout	Settings	Example
Single	n/a	One hero object.
Grid	Count x/y/z, Spacing	Speaker stacks, cube cities.

Radial	Count, Radius. Copies face outward.	Panels around an arena, crystal skyline.
Scatter	Count, Radius, Only surface, Seed	Floating debris, asteroid shells.
Orbit swarm	Count, Radius, Spread, Orbits / loop, Seed	Debris orbiting a planet.
Big swarm	Form (Orbits, Cloud, Shell, Galaxy), Count up to 250,000, Radius, Spread, Turns / loop, Seed	Huge swarms, placed by the graphics card on desktop and in WebGPU browsers (in WebGL2 browsers the processor places them, the same picture but slower, so keep counts lower there). Use a simple shape such as a tetrahedron. The **Glass Galaxy** preset has 40,000.
Wall	Columns, Rows, Spacing, Curve	Orb walls, LED walls. Curve bends the wall into an arc.
Spiral	Count, Radius, Height, Turns	Chandeliers, DNA helices.
Along a curve	Curve and Frequencies (the ribbon curves), Size, Count, Laps / loop, Face along	Beads riding a neon ribbon (give both the same curve, size and position), trains, conga lines.
On a shape's surface	On shape, Shape size, Count, Seed, Stand up, Lift. Copies are spread evenly by area.	Spikes on a crystal, moss on rocks. Set **Shape size** to the other layer's Size and put both layers in the same place.
On a terrain	Terrain (a Terrain layer), Count, Seed, Follow the slope, Lift	Trees, crystals, lamps and ruins standing on the landscape. They ride along as it scrolls, follow its placement and shrink away at its edges where it wraps; on water or lava they float.

Flock	Count (up to 2,000), Seed, Speed, Area, Formation; Spacing, Sight, Keep apart, Fly together, Stay close, Agility, Bank; Target x/y/z, Along a curve, Scatter; Close by, Tail, Steer back, Warm-up loops	Starlings, bats, fish, drones. The copies are **simulated**: each one steers by its neighbours and heads for the target (animate it, link it to the music or send it along a curve to lead the flock); link **Scatter** to kicks to make them burst outward on hits. The flight is worked out ahead of time into a loop. While it is (a few seconds, shown as *Simulating...* in the viewport bar) the flock shows its previous version or nothing, and exports wait for it. A free flock never repeats, so the end of the loop has to steer every boid back to where it started; **Formation** gives each boid a loose place of its own so the flight nearly repeats by itself and the catch-up is gentle. Under the settings, *seam* says how far the boids still are from their start when the final blend takes over. Use a small, flat shape (a squashed pyramid). The **Starling Dusk** preset has 600.

Physics	Scene (Rain, Stack and blast), Collide as (Box, Ball), Collider size; Rain: Count, Life, Area, Height, Spin, Leave by; Stack: Stack x/y/z, Gap, Blast at beat, Blast, Blast from; Gravity, Floor, Bounce, Grip, Seed; closing the loop	Copies as **solid bodies** that fall, stack and knock each other about, simulated ahead of time like flocks. **Rain** drops them one after another from the same places every loop; they pile up on the floor and shrink or sink away at the end of their **Life**. Keep the life short (under half of the loop-closing tail) and the loop closes without anything visibly sliding back into place. **Stack and blast** builds a wall or tower that a blast knocks down on the beat you choose; with ping-pong it builds itself back up in the second half of the loop, and with **Gravity** 0 the blocks burst apart in space. **Collider size** is half the collider for a copy of size 1: 0.5 for the built-in cube, 1 for the sphere. Random tilt and size under Variation don't change how copies collide, so keep them at 0. Up to 400 bodies. The **Beat Demolition** preset has a wall and a rain of pearls.

Liquid	Droplets, Container (Box, Bowl, Pool on the floor), Container size, Droplet size, Source (Filled, Poured in: Life, Spout height), Gravity, Viscosity, Cohesion, Stir, Seed, Liquid surface; closing the loop	Copies as the **droplets of a liquid**, simulated ahead of time like flocks. It fills a container around the layer's position (a box or bowl of the container's size, or a round pool on the floor) and settles; animate the layer's **Tilt** (Transform) and it sloshes from side to side. Put a **Bowl** shape on another layer at the same position, with the same Tilt and a Size of the container size ÷ 0.94, and the liquid rocks inside it. **Poured in** sends a stream from a spout above, each droplet living its **Life**. **Viscosity** makes it syrupy, **Cohesion** holds drops together, **Stir** swirls it (link it to the music). Droplets are copies of the layer's shape (small spheres, or glowing sprites on a sprite layer), sized by the layer's Size. Tick **Liquid surface** to draw them as one smooth liquid in the layer's material instead: Gold for molten metal, Glass for water (one liquid surface per scene; the floor's reflection shows its droplets). The loop closes by cross-fading two copies half a loop apart. A few thousand droplets bake in a few seconds; up to 8,000. The **Liquid Gold** preset rocks a bowl of molten gold.

Variation (appears when there are copies) adds life:

- **Random tilt**, **Random size**, **Random hue** and **Random spin** make every copy a little different. Change **Seed** to reshuffle them.
- **Size wave** is a wave of growing and shrinking that travels across the copies.
- **Light chase** is a bright glow running along the copies, like marquee lights.
- **Waves / loop** sets how many times those waves pass per loop, and **Wavelengths** sets how many waves fit across all copies.

Colours across copies (also when there are copies) spreads 2–4 colours over them: **Gradient** blends smoothly (back to the first colour at the end), **Steps** gives each copy one colour in turn. **Travel / loop** makes the colours run along the copies; **Colour the glow** tints the glow too.

Deform

Bends the shape itself, on every copy. All values have **~**, so they can move with the beat.

Setting	Meaning
Twist	Turns of twist from the bottom of the shape to its top.
Bend	Curves the shape into an arc (degrees from bottom to top).
Taper	Top wider (+) or narrower (−) than the bottom.
Wobble	Bumps flowing over the surface; **Bump size** and **Flow / loop** tune them. Simple shapes need **Subdivide** (in Relief) for smooth bends and bumps.
Explode	Faces fly apart, clearest with flat shading. Try **~** → Exp fade out, once per beat.

Placement & motion

Setting	Meaning
Position, Rotation	Where the layer sits and how it is turned (degrees).
Size	The size of the object, or of **each copy**. It does not change the distances between copies.
Stretch	Stretch per axis (make a cube into a slab or a pillar).
Spin / loop	Whole turns per loop around x, y and z. Negative numbers spin the other way.
Bob	Up/down offset. Animate it with **~** to make objects float.
Tilt	Rocks the layer side to side (degrees, around the depth axis). Animate it with a sine to rock a bowl, and give a liquid layer the same Tilt so it sloshes inside.
Animate on	Holds the layer's motion at a lower frame rate, like stop-motion or games animating "on twos": **Smooth** (the default) or 30, 24, 15, 12, 10, 8, 6 or 4 frames a second (or your own). Its spin, bob, shake and tilt, how its copies move (orbits, swarms, flocks, simulations), deform and glitch, and sprite-sheet frames jump from step to step, while the camera, texture scrolling, glow and colour pulses and particles stay smooth. So the loop stays seamless, the rate is snapped to a whole number of steps per loop; the line under it shows the steps and the exact rate on this loop (at 120 bpm and 16 beats, 12 fps is 96 steps). The **Stop-Motion Shelf** preset animates toys on 12, 8 and 6 fps.

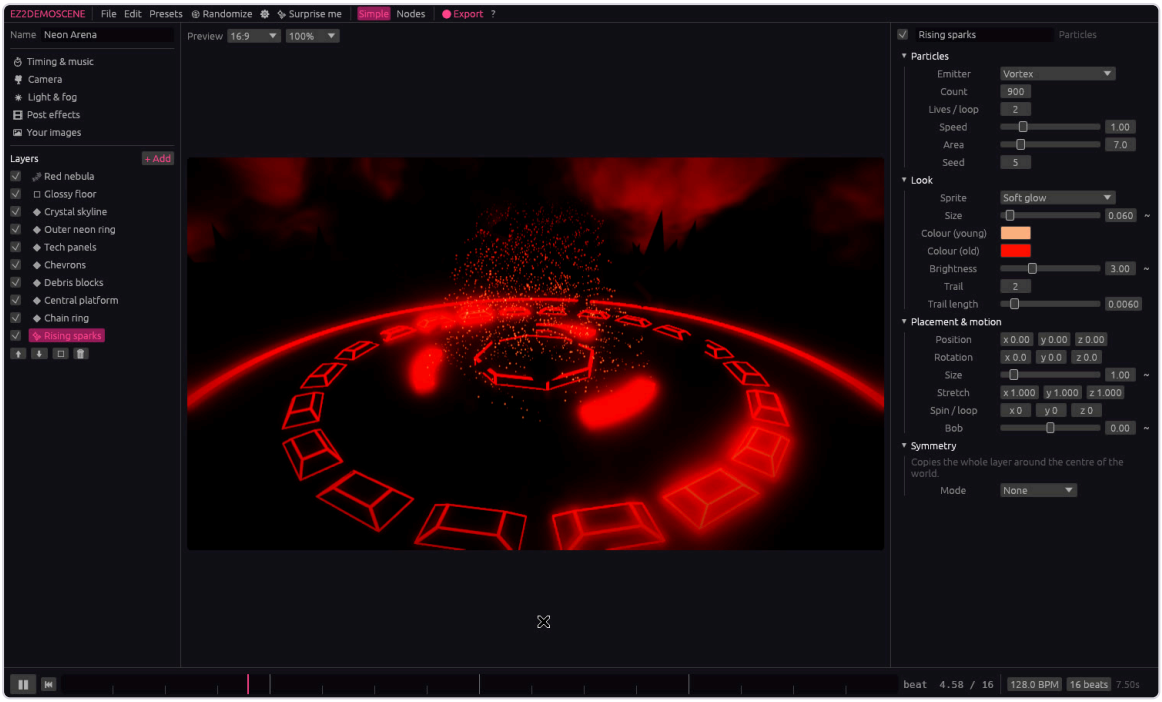
5.4 Symmetry

Symmetry copies the **whole layer** around the centre of the world:

- **Mirror X** (left/right), **Mirror Z** (front/back), **Mirror XZ** (four copies).
- **Radial**: N rotated copies around the vertical axis.
- **Kaleido**: N copies where every other one is mirrored, giving a kaleidoscope pattern.

> **Tip**
> Build one side of a room, then add **Mirror X**. The Gold Kaleido Room's orb walls, speakers and light panels are each built once and mirrored.

5.5 Particles



A particle layer.

Emitter	Motion
Burst	Explodes outward from the centre.
Sphere drift	Floats gently inside a sphere.
Ring orbit	Circles in a flat ring (Speed = whole orbits per loop).
Fountain	Shoots up and falls back.
Warp stars	Streams towards the viewer (hyperspace). Add a **Trail** for streaks.
Vortex	Spirals upwards like embers.
Snow / glitter	Falls slowly and sparkles.
Tornado	Spirals up a swaying funnel that widens with height, with debris circling its foot. Use it with **Smoke** for a dark twister, or bright blue for a water spout.

Smoke makes particles cover what is behind them instead of adding light, so they can be dark (smoke, dust, a tornado). **Brightness** then becomes **Opacity**.

Count is the number of particles. **Lives / loop** is how many times each particle is reborn in one loop, so higher means faster. **Area** is the size of the emitter. The look is controlled by **Sprite** (soft glow, pixel square, star, ring, or **Solid square (Quake)**: hard, unsmoothed squares of one colour that cover what is behind them, like Quake's sparks and blood), **Size**, two colours (young → old), **Brightness** and **Trail** (ghost copies behind each particle). **Flicker (light style)** makes the brightness step by a light style.

> **Performance**
>
> Particles are cheap. Tens of thousands are fine on a normal GPU. Trails multiply the count, so *count × (trail + 1)* is what gets drawn.

5.6 Terrain

An endless landscape that rushes towards the camera, in the retro vector style. Add it with **+ Add → 🗻 Terrain**.

Setting	Meaning
Style	**Wireframe** (glowing grid lines only; things behind show through), **Solid** (lit ground) or **Solid + lines**.
Height, Hills, Roughness	How tall the mountains are, how many hills fit across the terrain, and how bumpy they are.
Valley	Flattens a road down the middle (0 = none), the classic synthwave look.
Scroll / loop	How many times the landscape scrolls past per loop. It repeats exactly, so the loop stays seamless. 0 stands still.
Line colour, Line glow, Ground colour	The look. Animate Line glow with a Pulse for beat flashes.
Texture, Tiles, Texture on lines, Chunky pixels	Maps a texture onto the ground (and optionally the grid lines). It moves with the landscape; **Tiles** is how many times it repeats across the terrain, a whole number so the loop stays seamless.
Size, Grid cells	Its extent and resolution. More cells is smoother but slower. Grid lines sit on every cell.
Level of detail	Full resolution near the camera, gradually coarser further away, with no visible steps: a quarter of the triangles. Raise **Grid cells** (up to 512) or **Size** for sharper or bigger landscapes at the same cost. Distant ridges get slightly simpler.
Shape	**Hills** (rolling), **Ridged mountains** (sharp crests), **Mesas** (flat terraces with cliffs), **Sand dunes** (gentle on one side, steep on the other), **Canyons** (a plateau cut by winding gorges) and **Craters**.
Biome	Colours solid ground by height and steepness instead of one **Ground colour**: **Alpine** (grass, rock, snowy peaks, sandy shores), **Desert** (sand and banded red cliffs), **Volcanic** (black basalt with glowing cracks), **Arctic** (snow and blue ice), **Alien** (purple moss, teal crystal, glowing tips).

Water & lava

A **Liquid** fills everything below its **Level** with a flat surface. The level is a fraction of the mountain height; animate it for tides or rising lava. Liquids also show up on wireframe terrain.

Liquid	Look	Shine / Glow, Ripples / Crust…

Water	Reflects the sky with a sun glint, is lighter in the shallows and has foam and wet sand at the shore.	Shine, Ripples
Lava	Glowing rivers under a dark, cracked crust that drifts and churns. It lights up the nearby rocks.	Glow, Crust
Toxic goo	Glowing green slime with popping bubbles.	Glow, Bubbles
Ice	A frozen, cracked, shiny surface that stays still.	Shine, Cracks

Current / loop makes the surface drift along the terrain a whole number of times per loop; the ripples and crust also turn gently once per bar. Both loop exactly.

Turbulence adds Quake's wobbling liquid: the surface patterns sway by a sine of themselves, so the water, lava or slime seems to churn in place. The amount is in pattern cells (0.1–0.3 is Quake-like; animatable) and **Wobbles / loop** is a whole number, so it loops.

5.7 Laser beams

Club-style beams shooting from the layer's position along its up axis. Turn the layer with **Rotation** to aim them, and use **Mirror X** symmetry for a pair of laser rigs.

Setting	Meaning
Pattern	**Fan** (a flat fan that sweeps side to side), **Rotating cone** or **Scatter** (random directions that wobble).
Beams, Spread, Length, Width	How many beams, how wide the fan or cone opens (degrees), and their size.
Colour (first/last)	The beams blend from the first colour to the last.
Sweep, Sweeps / loop	How far the beams swing, and how often per loop (for the cone, also how fast it turns).
Beat strobe	Makes the beams flash on every beat and fade until the next one.
Style	**Laser beams** or **Spotlight cones**: wide, hazy cones of light, brightest where you look through the most haze. Set their **Cone angle**, and tick **Light pools** for bright spots where the cones hit the ground (height 0). Point the layer down (Rotation x 180) to hang the lamps from a ceiling. Add some **Mist** for a club look.

5.8 Neon ribbon

A glowing tube along a closed curve, with pulses of light running along it. Choose a **Curve** (**Lissajous**, **Knot**, **Figure eight**, **Wavy ring**, **Rose**) and its wave counts, then set **Pulses**, **Laps / loop** (negative runs backwards), **Pulse length** and **Pulse glow**. Ribbons reflect in mirror floors and can take symmetry.

5.9 Weather

+ Add → ☂ Weather fills a box around the camera with rain, snow, embers, sand or fireflies. The box follows the camera, so the weather is everywhere you look. Its **Ground height** is where rain

splashes and embers start.

Setting	Meaning
Kind	**Rain** (streaks and splash rings), **Snow** (swaying flakes), **Rising embers** (flickering sparks), **Sandstorm** (dust blown along the wind), **Fireflies** (wandering, blinking lights) or **None** (lightning only).
Amount, Size, Colour, Brightness	How many drops and how they look. Rain also has **Streak length** and **Splashes**.
Falls / loop	How many times each drop falls per loop: higher is faster. It is a whole number, so the weather loops.
Wind, Wind direction	Slants the rain and snow. Animate Wind with a Sine for gusts.
Area, Height	The size of the box around the camera.
Wet ground / Snow cover	Rain makes every surface darker and glossier, with puddles on flat ground that mirror the sky and ripple with rain rings. Snow settles on everything facing up. A snow cover can't really build up over a loop (it would have to jump back at the end), but you can animate it, for example with a **Swell** so it builds up and melts away.
Lightning	Jagged bolts that strike in front of the camera and a flash that lights up the whole scene, clouds included. **Chances / loop** and **Chance** decide how often; each strike flickers two or three times.

> **Performance**
> Weather drops cost about as much as particles. Volumetric clouds are the heavy part of a storm scene: lower their **Quality** on phones.

5.10 Waterfall

+ Add → 🌊 Waterfall adds a curtain of **water**, **lava** or **toxic goo** that pours from the layer's position downwards and arcs out along its Z axis, with streaks running down it and foam, spray or smoke at its foot. Place it on the edge of a cliff: a terrain mesa, a shape or a wall of boulders (see the Rainbow Falls preset).

Setting	Meaning
Width, Height, Arc	The size of the curtain and how far it arcs out from the edge.
Flow / loop	How many times the streaks run down per loop (a whole number, so it loops).
Brightness / Glow	Water brightness, or the glow of lava and goo.
Foam & mist / Smoke	Puffs at the foot (0 = none).
Turbulence	Quake's wobbling warp of the streaks (0 = none; animatable), one wobble per streak run.

5.11 Text

Glowing letters in the scene, on a flat sign facing forward (tick **Face the camera** to keep them readable from any side). They stay sharp at any size. Add one from **+ Add → 🔤 Text**.

Style	What it does
Still	The text, centred. Several lines are fine.
Scroller	Runs across a **Window** a whole number of times per loop (**Runs / loop**; negative runs the other way). Letters fade at the window's edges.
Sine scroller	A scroller riding a sine wave: **Wave height**, **Wave length** (letters per wave) and **Wave rolls / loop**.
Typewriter	Letters appear one by one, **Letters / beat**, starting again with the loop.
Greetings list	One line at a time, fading in and out, a new line every **Beats / line** (4 = one per bar).

Font & look: Pixel (chunky demoscene letters), Mono or Sans, or your own TTF/OTF file; size and spacing; a top and a bottom colour; **Glow** (brightness, above 1 it blooms; it has **~**); an outline; a drop shadow; and **Chrome**, which bevels the letters and reflects the sky like a 90s logo. The **Oldschool Intro** preset uses all three styles.

Numbers in text. Write `{0}` in the text and press **+ Number {0}**: the number shows there, and it has **~**, so it can count. Add more for `{1}`, `{2}`… Each number sets its fewest **digits** (padded with zeros: 007), its **decimals** and **1,000** (commas between thousands). Numbers are rounded down, so a count only shows its final value once it gets there. Examples: `TIME {0}` with a **once** ramp of −10 over 10 beats counts a timer down; `SCORE {0}` ramping by 987650 counts a score up; `COMBO {0}` can follow the music. Logos take numbers too.

5.12 Sprites

Pictures in the scene, one per copy: glows, flames, sparkles, coins, a logo on a sign. Add them from **+ Add → 🖼 Sprites** (Glow dots, Flames, Explosion or Image plane).

Setting	What it does
Image	A built-in texture, one of your images, or none (a soft glowing dot). The built-in **sheet_…** images are animated sprite sheets: explosion (plays once), flame, spinning coin and sparkle (these loop).
Facing	**Face the camera** (always square on), **Upright** (stands up and only turns around, like trees or flames) or **Fixed plane** (a flat picture you place and turn like any layer).
Blend	**Alpha**: soft edges from the image's transparency (copies are drawn far to near). **Additive**: light adds up, for glows and fire. **Cutout**: hard edges at half transparency, solid like a shape (best for pixel art). **Mesh (Saturn)**: a cutout with every other pixel left out in a checkerboard, the Sega Saturn's way of drawing see-through things (see Retro 3D).
Size, Opacity, Tint, Glow	Height (the width follows the picture), transparency, a colour multiplied in, and brightness (above 1 it blooms). All but Tint have **~** and 🎵.

Flicker (light style)	The glow steps by a light style: torches, candles, strobes.
Animation	For a sprite sheet (frames in a grid, left to right, top to bottom): **Columns**, **Rows**, **Frames** used (0 = all) and **Plays / loop**, a whole number so the animation lines up with the loop. **Each copy starts on its own frame** keeps a crowd of flames from flickering in step. Picking a built-in sheet or an animation (a GIF or video you imported) sets the grid for you, and an animation's Plays / loop so it runs as near its own speed as a whole number allows (**Play at its own speed** sets it again after you change the loop length).
Copies	The same copy layouts and variation as shapes (grid, ring, scatter, orbit, on a curve, on a terrain…).

The **Campfire Sprites** preset uses flames, sparkles and spinning pixel coins.

5.13 Electric arcs

Tesla-coil lightning, from **+ Add → ⚡ Electric arcs**. Each arc zigzags between its two ends, crawls while it lasts and takes a new shape at every strike.

Setting	What it does
Path	**Between two points** (**From** and **To**, placed with the layer), **To the nearest copies** of a shape or sprite layer (**Target**; from this layer's position to its **Arcs** nearest copies, so they jump as the copies move) or **Copy to copy** (from each copy of the target to the next, round the ring).
Strikes / loop	How many new shapes per loop, a whole number so the loop closes.
Jaggedness, Crawl, Fade	How far the arc zigzags; how far the zigzag slides during a strike; how much each strike dims before the next (0 = steady).
Branches, Width, Colour, Glow, Seed	Side forks, thickness, colour, brightness (**~** and 🎵: flash it on the kick) and a different set of zigzags.

The **Tesla Swarm** preset strikes the nearest debris from the core and runs arcs round a ring of coils.

5.14 Logos

A logo or title laid flat on the screen, over the whole scene: it stays the same size and in the same place whatever the camera does. Add one from **+ Add → 🏷 Logo** (Title, Chrome logo, Gold logo, Corner tag or Image logo). It is drawn before the post effects, so bloom, god rays, feedback trails, scanlines and the colour grade apply to it, but depth of field never blurs it.

Setting	What it does
Made of	**Text** (one or more lines, any built-in font or your own TTF/OTF, with numbers as in 5.11) or an **Image**. For an image, **Shape from** picks which parts are the logo: its **Transparency**, its **Bright parts** (a logo on black) or its **Dark parts** (a logo on white). Empty margins are cut off, so the size is the logo's own.

Attach to, Place	Like the reference points in a photo editor. **Attach to** is **The screen** or another logo layer. Click a square of the **Place** grid to snap there: on the screen the logo goes inside that corner, edge or the middle, a small margin in; against another logo it goes outside that side (the bottom square puts it underneath, the top right one at its top right corner, the middle one on top of it), with a small gap. An attached logo follows the other one wherever it moves, turns or grows. If the other logo is renamed or deleted, the logo goes back to the screen. **Attach to** also lists the **3D layers** (🧊): the logo is placed against the area that layer covers on the screen, and follows it as the camera and the layer move: a name under a spinning gem, a title riding above the moon, a label on each planet. While the layer is behind the camera the logo is hidden. The area is the box the shape fits in, so a label can sit a little away from a pointed or round shape: nudge it with the offsets.
Offset across, Offset up	Where the logo sits from that point, as fractions of the screen's width and height (right and up are positive). **Measured from** is the point the offsets start from (on the screen, changing it keeps the logo in place) and **Anchor** the point of the logo that sits there, which it also turns around.
Size, Turn, Opacity	Height as a fraction of the screen height (the width follows), a turn in degrees, and transparency.
Colours	**Gradient** (top and bottom colour) or **Image colours** (the picture's own colours times a **Tint**; for text, the tint).
Glow, Outline, Drop shadow, Chrome	The same looks as text: brightness (above 1 it glows), an outline ring, a shadow down and to the right, and a shiny bevel reflecting the sky.
Bevel	Under **Lighting**: gives the flat logo a relief lit by a light on the screen. **Round** edges, **Chiselled** (straight slopes, like cut metal), **Stepped** (terraces; set the **Steps**) or **Pillow** (the whole logo bulges). **Bevel width** is how far in from the edge it reaches, **Depth** how steep it is.
Light	**Light from** is the direction on the screen in degrees (0 from the right, 90 from above); ↻ **Circle the light** sends it round once per loop, and 🎵 can swing it on the beat. **Light height**: low lights rake across the bevel. **Shading** is how much the light darkens and brightens the colour, **Shine** and **Gloss** the highlights.
Material	A picture of a lit sphere (a *matcap*): each part of the bevel takes the colour of the sphere where it faces the same way. Built in: **matcap_gold**, **matcap_chrome**, **matcap_plastic** and **matcap_candy**, or import your own picture of a sphere. **matcap_environment** is chrome reflecting the scene's **Environment light** (a studio or panorama photo): it turns with the map and follows the camera, so the logo matches the shapes around it (without a map it looks like matcap_chrome). A material works without a bevel too, but only shows its middle colour.
Glint	A bright band sweeping across the logo, a whole number of **Sweeps / loop** (4 = every bar of a 16-beat loop), with its width, direction and colour.

Rings & outlines	**Rings**: glowing lines following the logo's shape, rippling outward a whole number of times per loop (negative: inward), fading with the **Reach**; optionally inside the letters too. **Stacked outlines**: solid outlines one outside the other, shading from the inner to the outer colour, like an 80s logo.
Extrude	Fake 3D: the logo repeated behind itself in a direction, fading darker, without any mesh.
Dissolve	Burns the logo away (0 whole, 1 gone) in patches, or from the edges inward, with a glowing burn front. Animate **Dissolve** with **~** to burn it in or out.
Reveal	How the logo appears as **Shown** goes from 0 to 1: **Grow from the middle** (the letters thicken from their middle lines), **Edges first** (outlines, then filling inward), a **Wipe** in any direction or a **Circle** opening from the middle, with a soft edge.
Morph	Melts the logo into another text (same font) or image: 0 is this logo, 1 the other. The shapes blend through their outlines, so a quick change reads best; a **~** sine with an amplitude above 0.5 holds each end and switches quickly. An image has its own **Shape from** (transparency, bright or dark parts), set apart from this logo's.
Copper bars	Amiga-style colour bars scrolling through the letters: two alternating colours, bright in the middle of each bar, a whole number of **Scrolls / loop**.
Wobble	**Sway** moves each row sideways and **Bob** each column up and down along a sine wave, rolling a whole number of times per loop: a rubber logo.
Glitch	Horizontal slices jump sideways with a colour split, with new jumps a set number of times per loop. Link **Glitch** to the kick with 🎵, or give it an Exp fade out on every beat.
Chromatic split	Red and blue pulled apart in a direction, like a misaligned CRT.
Pixel blocks	Under **Retro looks**: the logo in solid blocks (their size in logo heights). Animate it, for example an Exp fade out once per loop, to pixelate the logo in.
Palette	Only the colours of a retro machine (the same palettes as the post effect), with ordered **Dither**. **By brightness** uses the palette as a dark-to-light ramp, and **Cycles / loop** rotates the colours, like old palette-cycling animations.
Halftone	Printed dots, as big as the colour is bright, at a spacing and screen angle.
Scanlines, Phosphor stripes, Line glow	A CRT look on the logo only: dark gaps between lines, red-green-blue stripes and brighter lines that bloom.
Moiré	Two fine line patterns turning opposite ways over the letters, beating into shifting ripples.
Glass	Under **Glass, rays & echoes**: letters of glass. The scene behind shows through, bent by the letters' edges (their bevel, or a rounded edge without one), with **Dispersion** splitting the colours and a tint; the edges catch the light.
Rays	Light streaming out from the middle of the logo, a **Length** and a **Threshold**. With **Shadow** the rays are the light *behind* the logo, with the logo's shadow cut out of them: put it over something bright. Up to four logos can have rays.

Echoes	Fading copies of the logo where it was a moment ago, drawn behind it: set how many, the time between them (a fraction of the loop) and how quickly they fade. They show when the logo moves, turns or changes.

Every number has **~** and 🎵: pulse the glow on the kick, sway the turn, slide the logo in. With the logo selected, the viewport shows one handle at its anchor: drag it with **Move** to change the offsets, with **Scale** to size it and with **Rotate** to turn it (`Ctrl` snaps). The **C64 Intro** and **Copper Heaven** presets have logos with copper bars (the second with a sway), and **Glass Galaxy** has glass letters bending a galaxy, casting shadow rays through its light, with echoes as they sway.

> **Tip**
> The logo's shape is worked out once when you type or pick an image (it becomes a distance field, which keeps the edges sharp at any size and gives outlines, glows and bevels for free), so the first frame after a change can take a moment.

5.15 Mode 7 floor

The endless flat floor of SNES racing games (and the Saturn's VDP2 planes): a picture tiled across the plane at the layer's height, drawn per pixel right up to a hard horizon, with nothing underneath. Shapes stand on it and are hidden below it. Add it with **+ Add → 🏁 Mode 7 floor**.

Setting	What it does
Picture	Any built-in texture or your image; `track` (a race-track loop on grass) and `checker` are made for it.
Tile size	World units per tile (times the layer's **Size**).
Turns / loop	Whole turns per loop around the layer's position (put the position ahead of the camera for a racing turn, under it to spin in place). The layer's Y rotation turns it too.
Scroll / loop	Whole tiles per loop along x and z, so it seems to rush past.
Tint, Brightness	Colour multiplied in and brightness (animatable). The floor is not lit, like the console's.
Square pixels	No smoothing between the picture's pixels.
Fade into fog	Fades into the fog colour in the distance; off keeps the SNES's hard, bright horizon.

Its height is the layer's **Position** y. It isn't mirrored by the mirror floor and depth of field treats it as far away. With **Animate on** it turns and scrolls in steps. The **Mode 7 Circuit** preset races over a turning track at 256 × 224.

5.16 Gaussian splats

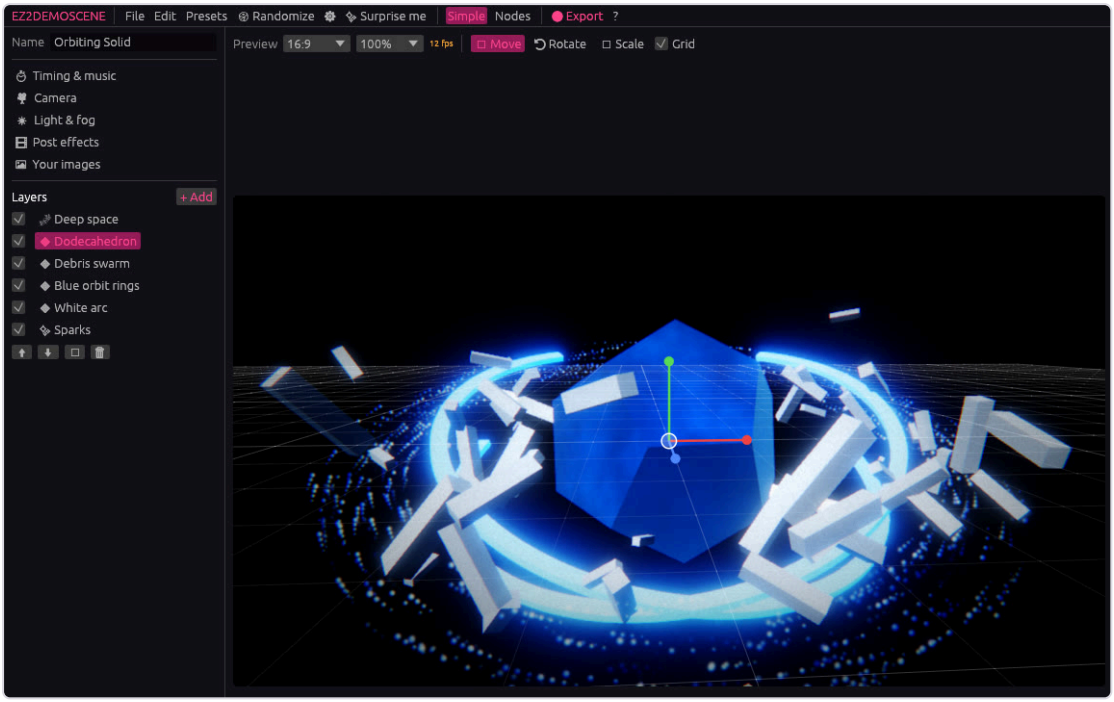
A captured object or place, made of millions of soft coloured blobs (**3D Gaussian splatting**), as phone apps and scanning tools make them. Add it with **+ Add → ☁ Gaussian splats** and choose its file, or open the file with **+ Add → 📦 3D model or splat file…** (or drop it on the window): `.ply` (as the training tools write it), `.spz` (Niantic's compressed format, about ten times smaller)

or `.splat`. A PLY file can hold a mesh or splats; its contents decide which layer it becomes. Without a file the layer shows a small built-in galaxy.

Setting	What it does
File	The splat file; **Built-in** goes back to the galaxy.
Up	Which way is up in the file. **As the format says** knows that PLY and .splat files keep their camera's upside-down frame and SPZ is Y up; pick another if the splats lie on their side.
Fit to size	Centres the splats and fits them into the layer's **Size**, like a model (stray splats far out are ignored). Off keeps the file's own units and origin: for a captured room or garden, turn it off and put the camera inside.
Tint, Brightness, Opacity	Colour multiplied in, brightness (above 1 glows) and opacity, all animatable.
Splat size	The size of every splat: 1 as captured, smaller turns the scene into dots (animatable: it dissolves and comes back).
Scatter	Moves every splat out from the centre; animate it for a scene that bursts apart and comes back together on the beat.
Most splats	The most drawn; above it the faintest and smallest are left out. Fewer is faster.

The layer's placement, spin, copies by **Symmetry**, blink and steps work as for any layer, so a captured statue can turn once per loop. Splats are see-through, drawn back to front after the solid shapes; they show in the mirror floor and fog, but cast no shadows, depth of field blurs them as much as what is behind them, and only the base colour of each splat is used (not the colours that change with the viewing angle). `.ksplat` and SOG files aren't read: convert them to PLY or SPZ.

5.17 Placing objects in the viewport



The Dodecahedron is selected: the Move handles appear on it, and the ground grid is on.

- **Select** a shape or particle layer by clicking it in the viewport, or in the layer list.
- Pick a tool above the viewport: **Move** (`W`), **Rotate** (`E`) or **Scale** (`R`).
- **Drag a handle**. Red is X, green is Y (up) and blue is Z. With Move, the white circle in the middle slides the object along the floor. With Rotate, drag one of the three rings. With Scale, the coloured squares stretch one axis and the middle circle changes the overall size.
- Hold `Ctrl` while dragging to **snap**: moves go in 0.25 steps, rotations in 15° steps and sizes in 0.1 steps.
- **Grid** (`G`) shows the ground grid, where one square is one unit.
- Dragging anywhere else still turns the camera, and the mouse wheel still zooms.

> **Tip**
>
> Pause first (`Space`) when placing spinning or orbiting things. Every handle drag counts as a single step for `Ctrl`+`Z`.

6 Materials, glow & textures

Surface

Presets sets up a ready-made physical material in one click: **Gold**, **Copper**, **Chrome**, **Brushed steel**, **Rubber**, **Car paint**, **Glass**, **Velvet** and **Ceramic**. They keep your textures and glow. The **Material Gallery** preset shows all nine.

Setting	Effect
Shading	**Classic** is the original look (and what older projects keep). **Physical** lights the surface the way real materials behave: sharper, truer highlights, reflections that get stronger at grazing angles, rough metals that stay bright, and three extra layers below. It shines under an **Environment light** (Light & fog), but works with the sky and ground colours too.
Colour	The base colour.
Metallic	0 looks like plastic or paint. 1 looks like metal, where reflections take on the colour (gold, chrome, copper).
Roughness	0 is a glossy mirror-like finish, 1 is matte. The glossy black look of Neon Arena is black + metallic 0.9 + roughness 0.15.
Faceted	Flat-shaded, low-poly look. Every face catches light separately.
Rim light	A glow along the silhouette that separates objects from dark backgrounds.
Clearcoat	(Physical) A clear lacquer on top, with its own **Coat roughness**: car paint, varnished wood, glazed ceramic. A second, sharp reflection over the colour.
Sheen	(Physical) A soft glow at grazing angles in the **Sheen colour**, like velvet or felt.
Glass	(Physical) Light passes through, tinted by the colour. **Refraction** is how much it bends (1.33 water, 1.5 glass, 2.4 diamond); a glass ball shows the world upside down, like a real one. Glass shows the environment behind it, not the other shapes.
Translucency	(Both shadings) Light shines through the surface from behind, like sunlight through a leaf, a paper lantern, wax, jade or an ear. Put the sun (Light & fog) behind the shape to see it. **Light inside** is the colour the light takes on passing through (warm orange for skin, yellow-green for leaves). The shape stays solid.
Transparency	(Both shadings) The shape is see-through: what is behind it shows, smoothly blended (0 = solid, 1 = invisible). Animate it with a fade to make shapes appear and disappear. Only the nearest surface of the shape is drawn, so its back and insides don't show through its front; for the Saturn's dithered see-through, use **See-through (mesh)** below.

Texture library

About 880 ready-made seamless textures, free to use for anything (CC0). Open it with 📚 **Texture library…** at the top of any texture menu, or with 📚 **Library…** next to the material presets. Two tabs, each with categories and a search box:

Tab	What is in it
PBR materials	135 photo-real surfaces by Poly Haven: brick, wood, bark, metal, rock, paving, roofing, fabric, concrete and plaster, ground. Each has a colour picture, a normal map and an occlusion/roughness/metal map (256 × 256). Picked for a shape's texture, it sets all three in one go and switches the material to Physical, with Colour white and Metallic and Roughness at 1 so the maps decide. Picked for the relief or the ORM map, it gives just that map; anywhere else (floors, terrains, backgrounds, sprites) its colour picture.
Low-res	740 small textures (at most 64 × 64, at most 128 colours) by Screaming Brain Studios and Kenney: brick, stone, wood, metal, floor tiles, roofs, walls, liquids, fabric, blocks and pixel patterns. Set the texture **Filter** to Nearest for crisp pixels.

> **Web app**
> The desktop apps carry the texture library; the web app downloads it (about 11 MB) the first time a library texture is needed. A project using library textures needs nothing else: packs don't copy them.

Material maps

Besides the colour **Texture**, two pictures fine-tune the surface. Both use the texture's Tiling and Scroll.

Map	What it does
Occlusion/rough/metal	One picture holding three maps, as glTF models and material sites ship them: **red** darkens creases (occlusion), **green** is roughness and **blue** metalness. They multiply the Roughness and Metallic sliders, so leave those at 1 to use the map as it is.
Glow map	Where the surface glows: it multiplies the glow colour (white glows fully, black not at all).

Glow (emission)

Glow colour and **Glow** (strength) make a surface emit light. Bright glow feeds the **bloom** post effect and produces the neon halo. **Glow where** decides which part glows:

Glow where	Result
Full	The whole surface glows (neon tubes, light panels).
Edges	Only the outlines of each face glow, for a Tron / wireframe-neon look. Works on the solids, panels, cylinders, shards and pyramids.

Stripes	Glowing horizontal stripes.
Texture	Bright parts of the texture glow (LED grids, circuits, tech-panel slits).

Glow flicker (light style) makes the glow step by a light style (a flickering neon sign, a strobing warning light).

Hue shift rotates all the colours of the layer. Animate it with a Saw wave for rainbow cycling.

See-through (mesh) makes a shape see-through the Sega Saturn way: instead of blending, a share of its pixels is simply left out in a fixed screen pattern. 0.5 is the Saturn's checkerboard "mesh" (ghosts, water, shadows); other values leave out more or fewer pixels, so animating it fades a shape in or out in dithered steps. With a low Retro 3D resolution the pattern is as chunky as the console's.

Morph

The **Morph** section of a shape layer melts its shape into another, like liquid: holes open and close, parts bud off and merge. Tick **Morph**, click **Into** to choose the other shape (any built-in shape, a library model or a file of your own) and set **Amount**: 0 is the layer's shape, 1 the other. Click **~** next to Amount to animate it, for example a sine that goes there and back once per loop, or a pulse on every beat.



A spaceship melting into a torus: Amount 0, 0.25, 0.5, 0.75 and 1.

Smooth or sharp

While Morph is on, the layer is raymarched from two distance fields (a grid of 64×64×64 around each shape): edges come out slightly rounded, parts thinner than a grid cell look ragged, and textures, relief, deform and glitch don't apply. Colour, metal, glow, rim light, copies and the colour ramp do. Untick it and the shape is the usual sharp mesh again. The first time a shape is used in a morph it takes a moment (about half a second on a phone) to build its field.

Relief: bump, normal and displacement maps

The **Relief** section of a shape layer makes its surface uneven. It uses the material's **Tiling** and **Scroll**, and the colour texture unless you pick a separate **Relief texture**.

Setting	Result
Mode: Bump	The texture's brightness is height: bright parts look raised, dark parts sunken (mortar between bricks, rivets, grid lines). Only the lighting changes, so it's cheap.

Mode: Normal map	The texture is a normal map (the purple-blue images from 3D tools and texture sites). Import one as an image and pick it as the relief texture.
Relief	Strength of the bump or normal map (animatable).
Displacement	Really moves the surface outwards by the texture brightness (negative pushes inwards). Animate it for pulsing, breathing shapes.
Subdivide	Adds geometry detail (each level splits every triangle in four), so displacement can show fine patterns.

> **Tip**
> Displacement looks best on smooth shapes (sphere, torus, capsule, rounded cube). Faceted shapes such as the cube or the gem open up at their edges, because each face moves along its own direction. 3D models (library or your own) are different: they stay closed, get the detail they need by themselves, and take textures from all sides. Tick **Faceted** in the material to light their bumps crisply.

Glitch

The **Glitch** section of a shape layer corrupts its geometry, like a broken signal. Set an **Amount** and a **Style**:

Style	Result
Jitter	The corners shake to random positions; the shape crumples without tearing.
Slices (VHS)	Horizontal bands of the object jump sideways.
Shatter	The faces fly apart along their directions.

Changes / loop is how many new random patterns appear per loop, and **Chance** is how many of those moments actually glitch (1 = all the time, 0.3 = short bursts). Like everything else, the glitch pattern repeats exactly every loop. Animate **Amount** with a Pulse for hits on the beat.

Textures

Pick a **Texture** from the built-in retro pack or from your own images. **Tiling** repeats it, **Scroll / loop** slides it by whole tiles per loop, and **Filter** sets how the texture is smoothed between its pixels:

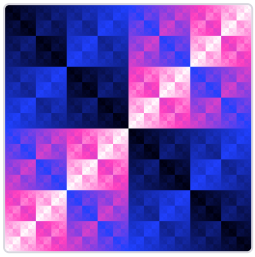
- **Smooth**: the modern look, with smaller copies of the texture further away so it never shimmers.
- **Nearest (PS1)**: square, sharp texture pixels, as on the PlayStation, the Saturn and in Quake (this was **Chunky pixels** in older versions).
- **Bilinear, no mipmaps**: blended between the four nearest pixels, but without the smaller copies, so fine patterns sparkle and crawl in the distance like early 3D cards.
- **3-point (N64)**: the Nintendo 64's cheaper blend of three pixels instead of four. Up close it is soft with a faint diagonal grain along the texture's pixel grid; it is what makes N64 textures look "smeary".

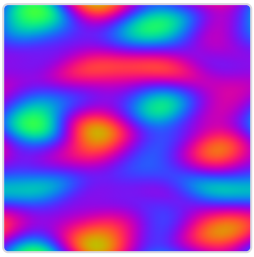
Turbulence makes the texture wobble like Quake's water, lava and slime: each coordinate sways by a sine of the other. The amount is in texture tiles (Quake: about 0.12; animatable), **Waves / tile**

how many wobbles cross a tile, and **Wobbles / loop** a whole number, so it loops. Pair it with **Glow where: Texture** for lava.

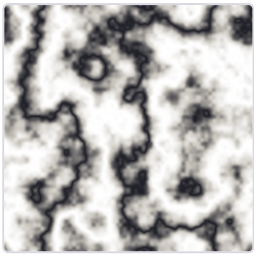
See Retro 3D for the rest of the 5th-generation look.

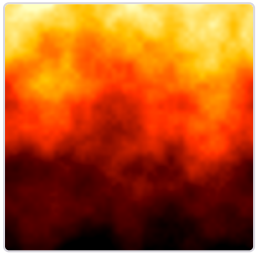

checker

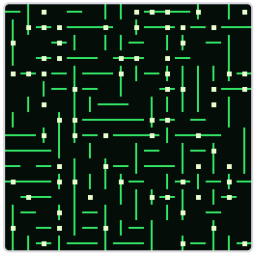

xor


plasma

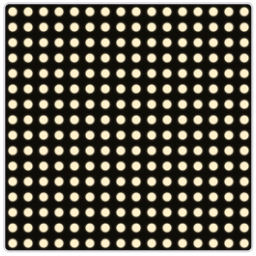

noise


marble

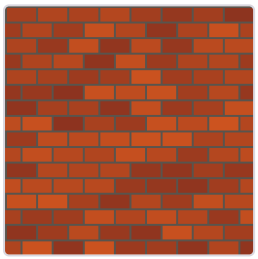

fire

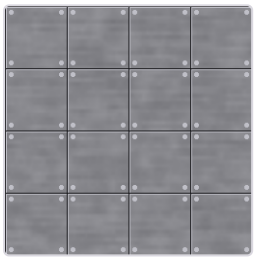

circuit

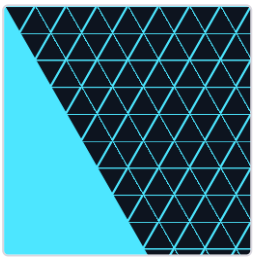

tech_panel


led_grid


speaker

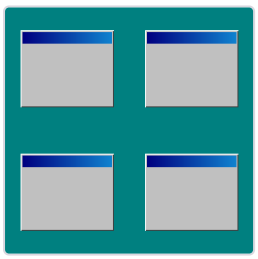

brick

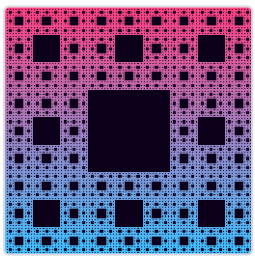

metal_plate


hex

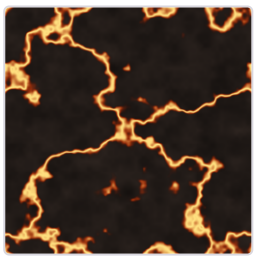

stripes

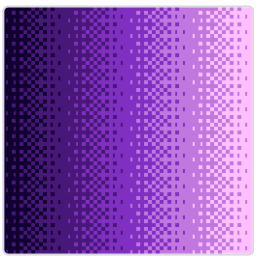

copper

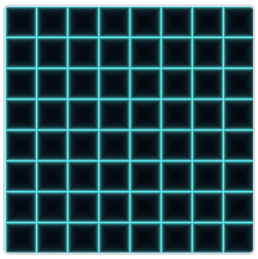

win9x

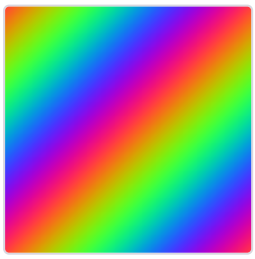

sierpinski

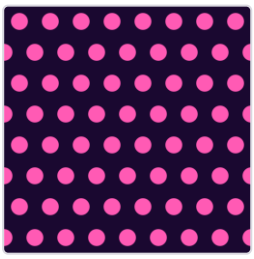

stars


lava

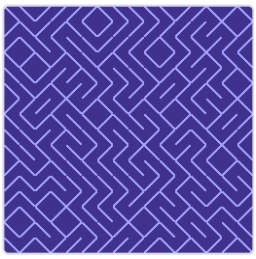

dither

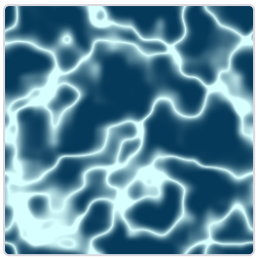

tron_grid

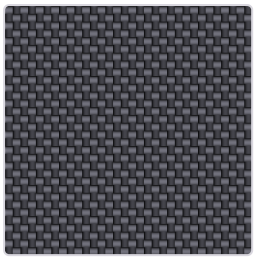

rainbow


polka

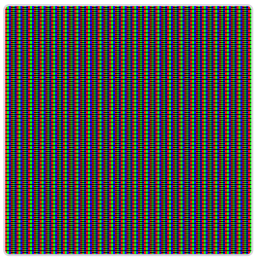

truchet

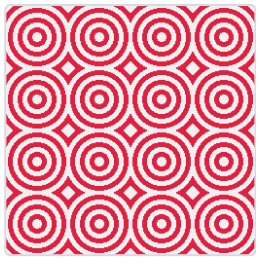

maze

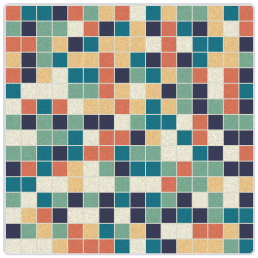

caustics


carbon

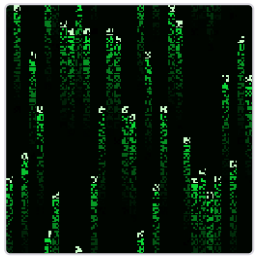

wood


crt_mask

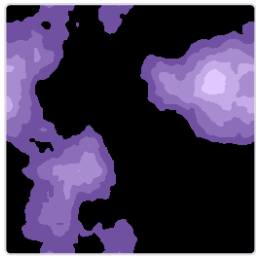

rings

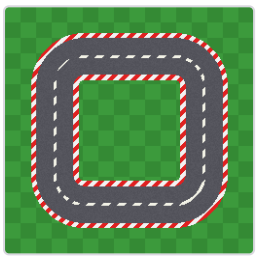

mosaic


clouds


matrix


sky_far


sky_near


track

Your own images & "retro-ize"

Import images (PNG, JPG, BMP, GIF, TGA), animated GIFs and videos in any of these ways: from a texture menu (**Import image…**), from **File → Import image…**, or by dragging them onto the window. A dropped image is applied to the selected shape. All your images are listed under **Your images**, where you can tick **Retro-ize**. This shrinks the image to **Max size** pixels and remaps it to a classic **Palette** with **Dithering**, turning any photo into a 90s texture.

A photo usually doesn't tile: where one copy meets the next, its left edge lands against its right edge and you see a seam. Tick **Mirror tiling** and every other copy is flipped, so edges always meet their own mirror image and the seams disappear. It applies everywhere the image is used (shapes, floors, terrain, the tunnel and battle backgrounds). A mirrored image repeats every two tiles, so when it scrolls, each scroll step moves it two tiles and the loop still closes.

Animated GIFs and videos

An animated GIF or a video (MP4, WebM, MOV, MKV…) imported like an image becomes an **animation**: its frames are laid out in one picture (a sprite sheet), which takes a moment while the status line says *Reading the frames…*. Up to 240 frames are kept, at most 320 pixels across; from a video, the first 30 seconds at up to 24 frames a second (fewer for longer videos). GIFs whose frames last different times are evened out.

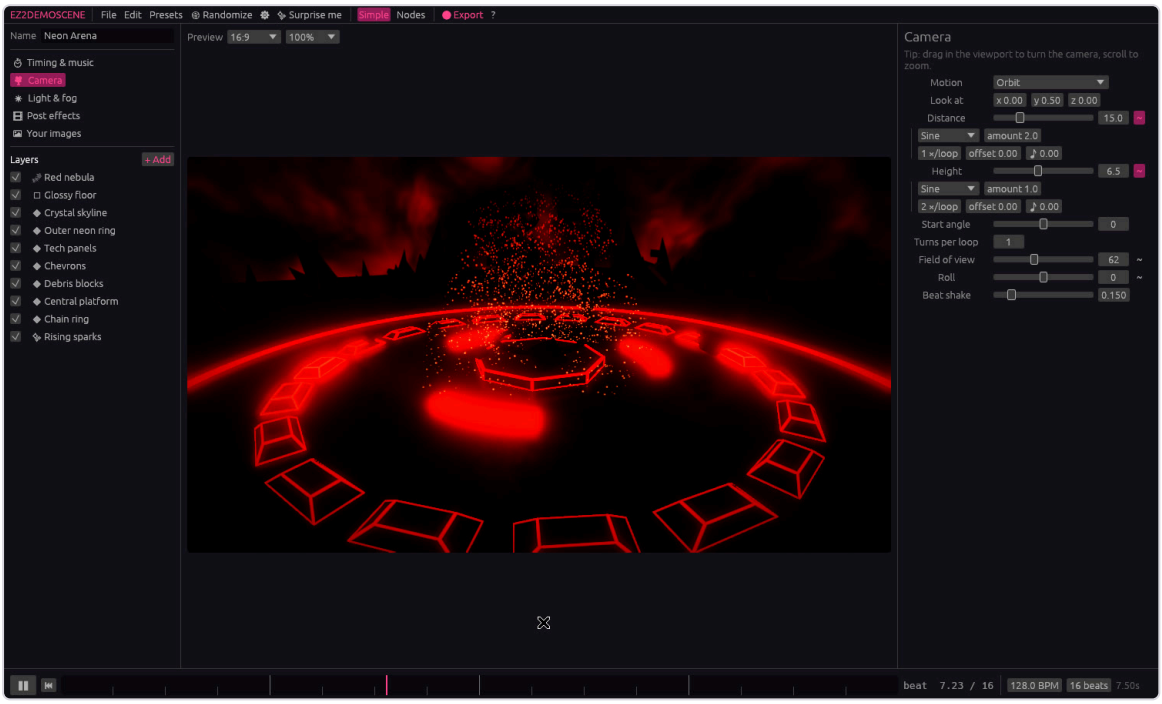
- In an **image layer** (Sprites: image plane, billboard…) it plays as a sprite sheet: the grid is set for you, and **Plays / loop** is the whole number of plays per loop nearest its own speed (a 2-second clip in an 8-second loop plays 4 times). Clips longer than the loop play faster to fit: make the loop longer to keep their speed.

- As any **other texture** (a shape, the floor, a terrain, a background) it plays by itself, the same way.

Your images shows each animation's frames, length and how it fits the loop. The desktop apps read videos with ffmpeg (bundled with the release downloads); the web app uses the browser's own video player, so a format plays if the browser plays it (MP4 and WebM in Chrome and on Android).

7 Animating anything

Every value with a ~ button next to it can move over time. The button turns pink when the value is animated. Click it to open the animation panel:



In the Camera section, Distance and Height are animated: their panels show wave, amount, ×/loop, offset and ♪.

Control	Meaning
Wave	The shape of the movement, in three groups (see below). Hover a shape in the menu for a description.
amount	How far the value moves from the slider value. 0 means not animated. Picking a shape or a rhythm on a still value gives it a starting amount so you see something straight away.
♩	Rhythm menu: **twice per beat**, **every beat**, **every 2 beats**, **every bar**, **every 2 or 4 bars** or **once per loop**. It sets ×/loop for you from the loop length.
×/loop	Whole cycles per loop. The number of beats in the loop hits **every beat**. Negative numbers run backwards.
offset	Shifts the timing (0–1 of a cycle), to stagger several animations.
♪	Adds the music's loudness (see chapter 10).

Below the controls, a small **graph** shows the value over the whole loop, with beat ticks and a dot at the current moment.

Group	Shapes	Use them for

LFO	Sine, Triangle, Saw up, Saw down, Square	Smooth or hard back-and-forth movement around the slider value.
Beat fades	Pulse (hit), Exp fade in, Exp fade out, Linear fade in, Linear fade out, Swell in & out	Envelopes that run from 0 to the full amount (or back) once per cycle. Set ♩ → **every beat** for hits that decay (Exp fade out), build-ups that drop on the beat (Exp fade in), or breathing swells.
Random	Sample & hold, Smooth random, Drunken walk	A new random value each cycle (held, or gliding smoothly), or a random walk that wanders and always finds its way back by the end of the loop.
Custom	Envelope	Your own shape over the cycle: points joined by curves, from the slider value (bottom) to the slider value plus the amount (top). See below.

Once: a change that happens one time

The last row of the panel, ⤴ **once by**, changes the value **once** and then holds it, instead of looping. Set how much it changes, **from beat** (when it starts), **over … beats** (how long it takes; 0 = at once) and the ease: **Linear**, **Ease in**, **Ease out**, **Ease in-out**, **Overshoot** (goes a little past, then settles), **Bounce** or **Jump at end**.

- In a **timeline** (chapter 10), the beats count from the start of the clip, so things happen when the scene comes in: menu items sliding in one after another (a logo's X starting off screen, **once by** the distance, each a little later), a title dropping in with Overshoot, a grade stamping down (Size shrinking with Overshoot while Opacity comes up).
- Outside a timeline, the beats count from the start of the loop, so the change plays again every loop and the loop point shows the jump back. Use it for things that are allowed to start over (a result screen), or put the scene in a timeline.
- It combines with the wave and the music: a value can slide in once and then pulse on the beat.

Envelope: your own shape

Pick **Envelope** and the graph becomes an editor for one cycle (with ♩ **once per loop**, the whole loop's timeline; with **every bar**, one bar that repeats):

- **Click** an empty spot to add a point (up to 64). Points snap to the grid; hold `Shift` to place them freely.
- **Drag** a point to move it in time and height.
- **Right-click** a point (long press on a touch screen) to choose how it goes on to the next point: **Straight**, **Smooth**, **Ease in**, **Ease out** or **Hold, then jump**, or to **delete** it. **Double-click** also deletes.
- The last point leads back round to the first, so the loop stays seamless. **offset** shifts the whole shape in time.



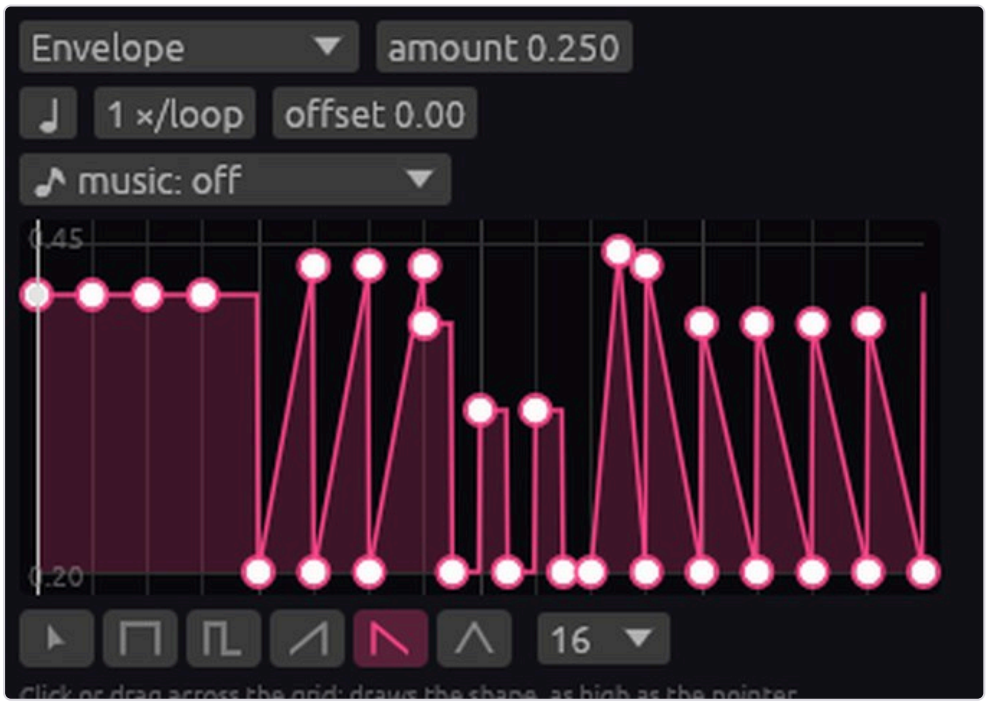
An envelope on Metallic: a rise, a hold, a peak and a slow fall over the loop. The labels are the real values at the bottom and top.

Drawing shapes

The buttons under the editor pick a tool. The arrow edits points as above. The others draw a shape into the cells of the grid, from the bottom up to the height of the pointer: click a cell, or drag across several to fill them all.

- **Step**: flat across the cell.
- **Square**: high for the first half of the cell, then low.
- **Ramp up**: rises across the cell, then drops.
- **Ramp down**: jumps up, then falls across the cell.
- **Triangle**: up and back down.

The number next to the tools sets the grid: **4, 8, 16 or 32 steps** in one cycle. With ♩ **once per loop** on a 16-beat loop, 16 steps is one per beat. Only the cell you draw in changes; the rest of the envelope stays as it was, with a jump at each edge where needed. Right-click an empty spot for **Clear** (flat at the bottom) to start again.



Drawn with the tools on a 16-step grid: steps, ramps up, squares, a triangle and ramps down.

Almost every number can be animated this way, including **post effects** (bloom threshold, pixel size, dither, CRT scanlines and curvature, contrast, saturation, vignette, grain, beat flash, kaleidoscope angle), materials (metallic, roughness, rim light, tiling), light, camera swing and shake, particle speed and area, mirror reflectivity and blur, laser spread, length, width and sweep, terrain roughness and valley, and ribbon pulses.

Recipes

- **Pulse on every beat:** Glow → Pulse, amount 3, ×/loop = loop beats (e.g. 16).
- **Breathing camera:** Camera Distance → Sine, amount 2, ×/loop 1.
- **Kick zoom:** Kaleidoscope Zoom → Pulse, amount 0.2, ×/loop 16.
- **Rainbow:** Hue shift → Saw, amount 0.5, ×/loop 1.
- **Floating object:** Bob → Sine, amount 0.5, ×/loop 2.
- **Strobe:** Glow → Square, ×/loop 32.
- **Beat hit:** Glow → Exp fade out, ♩ every beat.
- **Build-up:** Bloom threshold → Exp fade in, ♩ every 4 bars (negative amount lowers it towards the drop).
- **Nervous camera:** Camera Height → Drunken walk, ♩ every beat, amount 1.
- **Drop:** Bloom threshold → Envelope, ♩ once per loop: flat, a slow ease-in over the build-up, then a hold and a jump right on the drop.

Shake

Placement & motion → Shake jolts a layer in a random direction. **Shake distance** and **Shake turn** set how far, and **New direction** how many times per loop a new random direction is picked. Give the distance a fade for a hit that settles: the ⚡ **Beat jolt** button does exactly that (Exp fade out, every beat). The **Jitter** node can drive the same shake for everything connected to it (chapter 12).

Blink / strobe

Every layer has a **Blink / strobe** section that makes the **whole layer** appear and disappear, or flash:

Mode	What it does
Blink	On and off in a regular rhythm. **On time** is the fraction of each blink it stays visible.
Random blink	On and off at random moments. **Chance on** sets how often it is visible.
Flash	Stays visible but its glow flashes and fades. **Dim level** is the brightness between flashes.

Times / loop sets the rhythm (16 = every beat of a 16-beat loop) and **Offset** shifts it, so two layers can blink in turn.

8 Camera, light & fog

Camera

Setting	Meaning
Motion	**Orbit** circles the target **Turns per loop** times. **Pendulum** swings back and forth by **Swing** degrees. **Static** stays put, though its values can still be animated. **Path** flies through your own points (see *Camera paths* below).
Look at	The point the camera aims at.
Distance, Height	How far away and how high the camera is. Both are animatable.
Start angle	Which side the camera starts from.
Field of view	Zoom lens: 20–35 for a telephoto look, 60–90 for wide and dramatic.
Roll	Tilts the horizon. Animate it for a dizzy rave feel.
Beat shake	A small camera kick on every beat.
Punch-in	With a song loaded, the camera zooms in on every kick (or snare, hi-hat, any hit, note) and eases back.

> **Mouse**
> Dragging in the viewport changes **Start angle** and **Height**. Scrolling changes **Distance**. A Path camera flies by itself, so dragging does nothing then.

Camera paths

Pick **Motion → Path** and the camera flies a smooth closed loop through your points, at an even speed, **Laps / loop** times per loop. Each point has a position (**Eye**), a **Look at** point, a field of view and a roll, and the camera blends between them. With the Camera selected, the viewport shows the flight as a yellow line with numbered points.

- **Adding shots:** switch to **Static**, frame the shot by dragging and scrolling in the viewport, then press 📍 **Add this view to the path**. Repeat, then switch back to **Path**.
- **Changing a shot:** press 👁 on a point to look from it (the camera switches to Static there), adjust it in the viewport, then press ⟳ on that point to store the new view. ⬆ ⬇ reorder points, 🗑 removes one.
- **Linger** slows the camera down at every point: 0 is an even glide, 1 holds each shot for a moment.
- **Cut on hits**: with a song loaded, the camera cuts to the next point on every kick (or snare, hi-hat…) and **Drift**s a little towards the following one. The cuts restart with the loop, so the video still loops. Without music the camera just flies.

Light & fog

- **Fog colour** / **Fog density**: distant things fade into this colour. The fog colour is also the "empty space" colour, so match it to your background.

- **Sky light** / **Ground light** / **Ambient**: soft light from above and below. The sky colour is also what glossy surfaces reflect, which gives black gloss its highlights.
- **Sun colour**, **Sun strength**, **Sun direction**: the main light.
- **Sun & ambient flicker (light style)**: the sun and the ambient light step by a light style, like a flickering room in Quake.
- **Environment light**: what surrounds the scene and lights it from every side, and what shiny things reflect. **Colours** (the default) uses the sky and ground light above. A **studio** (*Softbox*, *Overcast*, *Sunset*, *Neon room*) or a **panorama photo** (**Panorama photo (.hdr)…**, or drop a `.hdr` file on the window; free ones are on sites like Poly Haven) lights shapes like a photo of a real place: chrome shows the room, rough plastic picks up its colours softly. **From the sky** uses the scene's own background, so clouds and sunsets light the objects; tick **Capture once** for a background that doesn't move (it is cheaper). **Strength** brightens it, **Turn** spins it (↻ once per loop for a slow sweep of reflections). **Sun from the map** finds the brightest spot of the map (the sun in a photo) and makes it the sun, with its shadows, so they line up with the picture. Add the **Environment map** background to see the map behind the scene. Shapes, 3D models and raymarched shapes are lit this way; terrain, water and the mirror floor keep the classic light. The **Chrome Studio** preset has chrome, gold and plastic under a turning sunset.
- **Reflections**: shiny shapes, water and rain puddles reflect the things around them, not just the sky or the environment map: a chrome ball shows the red box beside it, a lake mirrors its islands. **Strength** is how much of that replaces the plain reflection, **Reach** how far away things can be (in world units), and **Up to roughness** which surfaces take part (rougher ones keep the plain reflection). It can only reflect what is on the screen, so towards the picture's edges and behind shapes it fades back to the environment. The mirror floor keeps its own exact reflection. It costs one extra half-resolution pass (shared with depth of field when both are on).
- **Light shafts**: the sun lights the fog wherever it reaches it, so shapes, copies and terrain cut dark bands through the haze, like sunbeams through a window or between trees. Unlike the **God rays** post effect, they work with the sun off screen, and they follow the day cycle. They need fog or mist (no fog, nothing to light) and the sun's shadows, which turning them on switches on. **Strength** is how brightly the fog glows, **Scattering** how much more it glows when you look towards the sun (0 the same all round), **Reach** how far into the fog the light is gathered, and **Quality** the number of samples along each view (fewer is faster, grainier). Beyond the shadows' **Distance** the fog counts as lit. The **Cathedral Light** preset shows them.
- **Sun shadows**: shapes and terrain cast shadows from the sun (the moon at night) onto shapes, terrain, water and the mirror floor. **Darkness**, **Softness** and **Distance** (how far around the camera's target shadows reach: smaller is sharper). In dark neon scenes shadows are subtle, because most light comes from glow, not the sun.
- **Contact shadows**: soft dark patches under shapes standing on a mirror floor, so they don't look like they float.
- **Mist (height fog)**: fog that is thick near the ground and thins out higher up, so it pools in valleys and over water. **Base height** is where it is densest, **Thickness** how high it reaches. It also hides the horizon of the background.
- **Underwater caustics**: the rippling web of light you see at the bottom of a pool, on every surface. **Only below** limits it to things under a water line.
- **Rainbow**: a rainbow (with a faint second bow) opposite the sun. Like a real one, it needs the sun fairly low: with the sun high above, the bow sinks below the horizon.
- **Day & night cycle**: the sun rises, crosses the sky and sets a whole number of times per loop. The light, fog and sky turn warm at sunrise and sunset; at night the sky darkens, stars come out,

the moon rises opposite the sun and moonlight takes over. **Sun direction** sets where the sun is at noon. Clouds, god rays and reflections all follow the sun.

Light styles (Quake)

A light style makes a light flicker by a string of letters, the way Quake's lights did: each letter is a brightness, from **a** (dark) through **m** (normal) to **z** (about twice as bright), and the letters play one after the other. Light styles work on the sun and ambient light (**Light & fog**), a material's glow, sprites and particles.

- **Style**: ready-made patterns: **Flicker**, **Candle**, **Torch**, **Pulse** (dark to twice as bright and back), **Slow pulse**, **Strobe**, **Slow strobe** and **Broken fluorescent**. **Steady** turns it off.
- **Letters**: type your own (only a–z count; e.g. `mmamammmmam` for a failing tube).
- **Plays / loop**: the whole string plays a whole number of times per loop, so the loop stays seamless. The panel shows the resulting speed in letters per second; **Quake speed** picks the number of plays closest to Quake's ten letters a second. Picking a style does the same.

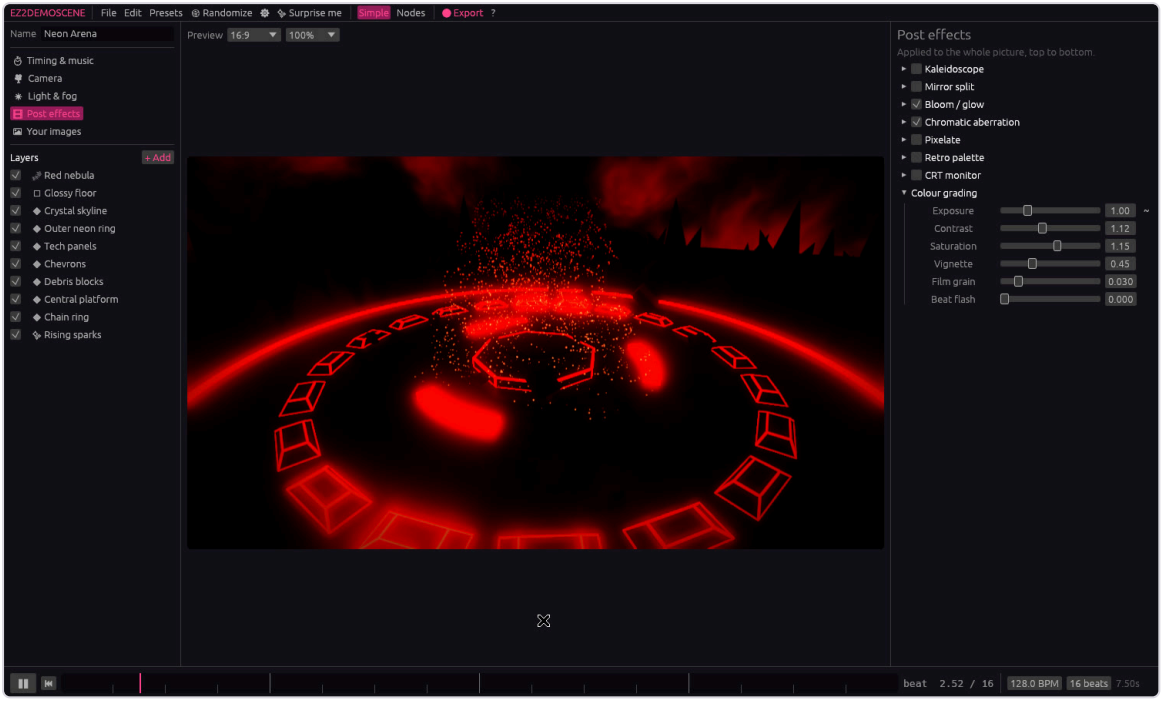
Colour scheme

🎨 **Colour scheme** (left panel) brings every colour of the project into harmony with one **key colour**. Each colour keeps its *lightness*, so darks stay dark and glows stay bright, while its hue moves to the nearest of the scheme's hues. The work happens in a perceptual colour space (OKLCH), so a colour's brightness doesn't change when its hue does. Your own colours are kept: turn the scheme off and they're back. Pictures (textures, sprites, imported images) keep their own colours; the **Retro palette** post effect is the way to force those into a palette.

Setting	What it does
Key colour, Harmony	The colour everything goes with, and which hues go with it: **One hue**, **Neighbours** (30° either side), **Opposites**, **Split opposites**, a **Triad** or a **Square**. The swatches show the scheme's hues.
Hue pull	How far colours move toward the scheme: 0 changes nothing, 1 puts every hue on it.
Saturation match	0 keeps each colour's own saturation; 1 makes them all as saturated as the key.
Tint greys	Greys and whites take the key's hue too. A new shape or 3D model starts grey, so this is what brings it into the scheme; 0 keeps greys neutral. Near-black stays dark either way.
Turn	Degrees added to the key's hue. Animate it (↻ **Turn once per loop**) and the whole scene goes round the colour wheel together.
Sky, fog, sun and rays too	Whether the environment follows the scheme.

Near-greys (black floors, white letters) are left alone. A layer whose colours mean something, like fire, can tick **Keep own colours** at the top of its settings (shown while a scheme is on). With a scheme on, 🎲 **Randomize** picks a new key colour (and sometimes a new harmony) instead of rotating each layer. The try it on **Neon Arena** with a triad and a turn of once per loop.

9 Post effects



Post effects. Tick a section's box to enable it, and click its arrow to open its settings.

Effect	What it does
Kaleidoscope	Folds the whole picture into mirrored wedges. You can set Segments, Angle, Turns per loop, Zoom (animatable) and Centre.
Mirror split	Mirrors one half of the picture onto the other (left → right, top → bottom, or quad). This is instant symmetry.
Bloom / glow	Bright things spill light. Intensity sets the strength, Threshold how bright something must be to glow, and Spread how wide the halo is.
God rays & lens flare	Shafts of light streaming from the sun (or from the picture centre, for "light at the end of the tunnel"). Bright sky near the light is smeared towards it, so clouds, mountains and shapes in front cast streaks of shadow. Length, Threshold and Tint shape the rays; **Lens flare** adds ghosts and a streak when you look into the light, and fades when the sun is hidden.
Depth of field	A camera lens blur: things nearer or further than the focus go soft, with round bokeh. **Auto focus** keeps the point the camera looks at sharp (or set a **Focus** distance); **Blur** sets how soft it gets. Both have ~ and 🎵, so the focus can pull or the blur can breathe on the kick.
Feedback trails	Each frame keeps a fading copy of the previous ones, so moving things leave echoes. **Trail length** (with ~ and 🎵) sets how long they linger; **Zoom / second** makes the echoes fly out (above 1) or fall in, **Turn / second** spins them and **Colour drift** changes their hue as they fade. The speeds are per second, so they look the same at any frame rate. The preview builds trails as it plays and starts fresh after a jump; exports render one loop first (not saved), so frame 0 already has its trails and the loop closes seamlessly.

Heat haze	The picture shimmers as if seen through rising hot air: above hot, bright things (lava, fire, the sun), near the bottom of the picture (hot ground) or everywhere. **Rises / loop** sets its speed.
Fisheye lens	**Amount** above 0 is a fisheye: the middle bulges out towards you and the corners stay where they are. Below 0 is the opposite: the middle shrinks away and the edges stretch. Animate it with ~ for a lens that breathes, or put it on the kick with 🎵.
Line wobble	The lines of the whole picture wobble like a retro RPG battle: **Wave** (lines slide sideways), **Interlaced** (odd and even lines slide opposite ways; **Lines** sets how fine) or **Compression** (lines squeeze up and down). **Amount**, **Waves** and **Waves / loop** as for the battle background.
VHS tape	A worn video tape: lines jitter, colour bleeds sideways, snow, faded colour, and a torn **Tracking band** rolling down the picture three times per loop.
ASCII art	The picture drawn with text characters, denser where it is brighter. **Rows** sets how many lines of text; **Colour** keeps the picture's colours or uses a green, amber or white terminal; **Picture behind** lets some of the picture show between the characters.
Chromatic aberration	An RGB colour split towards the edges, like a cheap lens. Pulse it on the beat.
Pixelate	Fat pixels (the size is measured at 1080p, so previews match exports).
Retro palette	Reduces the image to a classic computer palette with ordered (Bayer) dithering.
CRT monitor	Scanlines, screen curvature and a VHS wobble.
Colour grading	Exposure, Contrast, Saturation, Vignette (dark corners), Film grain and Beat flash (a white flash on every beat).



Neon Arena with an interlaced line wobble, a VHS tape and ASCII art in a green terminal.

Retro 3D: 5th-generation consoles

🕹 **Retro 3D** (left panel) draws the whole 3D scene the way the PlayStation, Saturn and Nintendo 64 did. Unlike the post effects above, it changes how shapes are drawn, not the finished picture. Click a **Look** to turn on a console's bundle of settings, then fine-tune it:

- **PlayStation**: 320 × 240, wobbling vertices, fully warped textures, square texture pixels, dithered 15-bit colour.
- **Saturn**: 320 × 224, wobbling vertices, milder warp, square texture pixels, 15-bit colour without dither. Add **See-through (mesh)** to materials or the **Mesh** blend to sprites for its transparency.

- **Nintendo 64**: 320 × 240, no wobble or warp (the N64 corrected perspective), soft 3-point filtered textures, fog starting close to the camera, dithered 15-bit colour smoothed by the video blur.
- **Quake (software)**: 320 × 200, square texture pixels and palette lighting through the 256-colour Software 3D palette with glowing fullbrights. Add light styles, turbulence, the two-layer sky and solid square particles for the rest.

Modern (off) resets it.

Setting	What it does
Whole screen (8- and 16-bit machines)	Draws the *whole* picture, text, logos and effects included, at an old machine's resolution: one colour per machine pixel, sampled after everything else, so pixels can be wider than tall. The **Machine** buttons set the size, shape, border and palette of the C64 (multicolour 160 × 200 with its double-wide pixels, or hi-res 320 × 200), ZX Spectrum, Amstrad CPC mode 0, CGA and VGA PCs, Amiga, NES, Master System, Mega Drive, Game Boy, Virtual Boy and the 1-bit Macintosh; the palette goes to **Post effects → Retro palette**, where you can change its dither. **Shape**: **4:3 television** (the machine's picture on a TV, with the border colour at the sides of a wide output), **Square pixels** (handhelds and the Mac) or **Fill the output**. **Border** is the colour around the picture (the C64's purple-blue, the Spectrum's blue…) and **Border size** how much of it there is inside the frame, at the sides and top and bottom: the Spectrum and C64 buttons set their wide borders all round. Top and bottom only makes a **letterbox**, like the smaller window Star Fox drew its 3D in (the **Super FX Starship** preset). **Loading stripes** turns the border into a tape loading, as on the ZX Spectrum: **Pilot tone** (broad red and cyan bands rolling), **Data** (thin, flickering blue and yellow bands) or **Loading screen**, the whole load over the loop: the pilot tone over a black picture, then the picture's lines arriving in black and white in the Spectrum's memory order (the top line of every character row of the top third, then the second line of each…), then its colours filling in by character rows, and a moment finished before it starts again. The **Tape Loader** preset shows it. **Bands** and **Roll / loop** set the pilot bands. With **CRT monitor** on, there is one scanline per machine line. Combine it with **Draw the scene at** for 3D that aliases at its own resolution underneath.
Draw the scene at	Draws the 3D scene small (**256 × 224**, **320 × 240**, **640 × 480** or your own size) without smoothing, then blows it up with square pixels. Edges stair-step, thin things break up and shading is blocky, as on the console: this is not the same as the **Pixelate** post effect, which only blocks up the finished, smooth picture. Sizes are for a 4:3 television; on a wider output the height is kept and the width grows, so the pixels keep their shape (320 × 240 at 16:9 is 427 × 240). The line under it shows the size at your export size. Post effects (bloom, CRT…) still run at full size on top.
Sharp text & logos	Text layers and logos stay at full resolution on top of the chunky scene (like a modern remaster's menus). Untick it to make them as chunky as the scene. Sharp text is still hidden behind shapes in front of it.
Snap vertices	The PlayStation had no fractions of a pixel for the corners of its triangles, so they jump from pixel to pixel: shapes shiver, wobble and crawl as they or the camera move. **Grid** is the pixel grid the corners snap to (usually the same as the resolution; a coarser grid wobbles more). **Amount** (animatable) fades the wobble in and out. Shapes, 3D models, terrain and sprites snap.

Each material's filter / Texture filter	Untick **Each material's filter** to give every shape and the terrain one texture **Filter** (see Textures), e.g. Nearest for the whole PlayStation look at once. Terrain has only square or smooth pixels.
Near-plane culling	Triangles with a corner closer to the camera than this distance (world units) vanish whole, as on the PlayStation, which dropped them instead of cutting them: walls and floors pop open when the camera gets close. Big polygons (a floor made of two triangles) vanish as soon as a corner is behind the camera, which is why games split them up: **Subdivide** them. 0 is off; animatable.
15-bit colour	Every shape, the terrain and sprites round their colours to 32 levels per channel as they are drawn, as the consoles' frame buffers stored them, so smooth shading turns into bands. This happens per polygon, before post effects, and is far subtler than the **Retro palette** post effect.
Dither	A fixed 4 × 4 pattern added before the rounding hides the bands in a fine grid of dots (the PlayStation's and N64's look; the Saturn didn't). 0..1, animatable. At a low resolution the dots are console pixels.
Palette lighting (Colormap)	Light steps through a palette's own colours, like Quake's software renderer: each surface colour (texture included) becomes its nearest palette colour, and shading picks a darker or brighter palette colour for it from a table of **Light levels** (Quake: 32) running from black to twice as bright. Shading turns into bands of palette colours rather than smooth gradients. The **Software 3D (256)** palette is our own: 14 ramps of 16 shades (greys, browns, stone, greens, reds, rust, gold, skin, purples, blues, teal, olive, clay, sky) and 32 **fullbright** colours (fire and lamp colours) that glow whatever the light (untick **Fullbrights** to light them too). Any retro palette works as well (EGA, C64, VGA cube…), without fullbrights. Shapes and terrain use it.
N64 fog	Instead of the scene's fog, fog that begins at **Starts at** (close to the camera) and thickens in a straight line to solid fog at **Solid at**, in the scene's fog colour, like the Nintendo 64's short draw distances. Height fog still adds on top. Both distances are animatable (roll the fog in).
Video blur	The Nintendo 64's video output filter: neighbouring pixels that differ by less than a dither step are averaged (dither dots melt away) and the picture is softened sideways, one console pixel wide. It is what makes N64 pictures look soft. 0..1, animatable.
Texture warp	0..1 (animatable). The PlayStation and Saturn stretched textures straight across each triangle on the screen instead of in perspective ("affine" mapping), so textures on big polygons bend and swim as the view changes, most of all on floors and walls seen at an angle. To make it smaller, do what PlayStation games did: split the shape into smaller triangles with **Subdivide** (in **Shape**); each level splits every triangle in four and roughly quarters the warp.

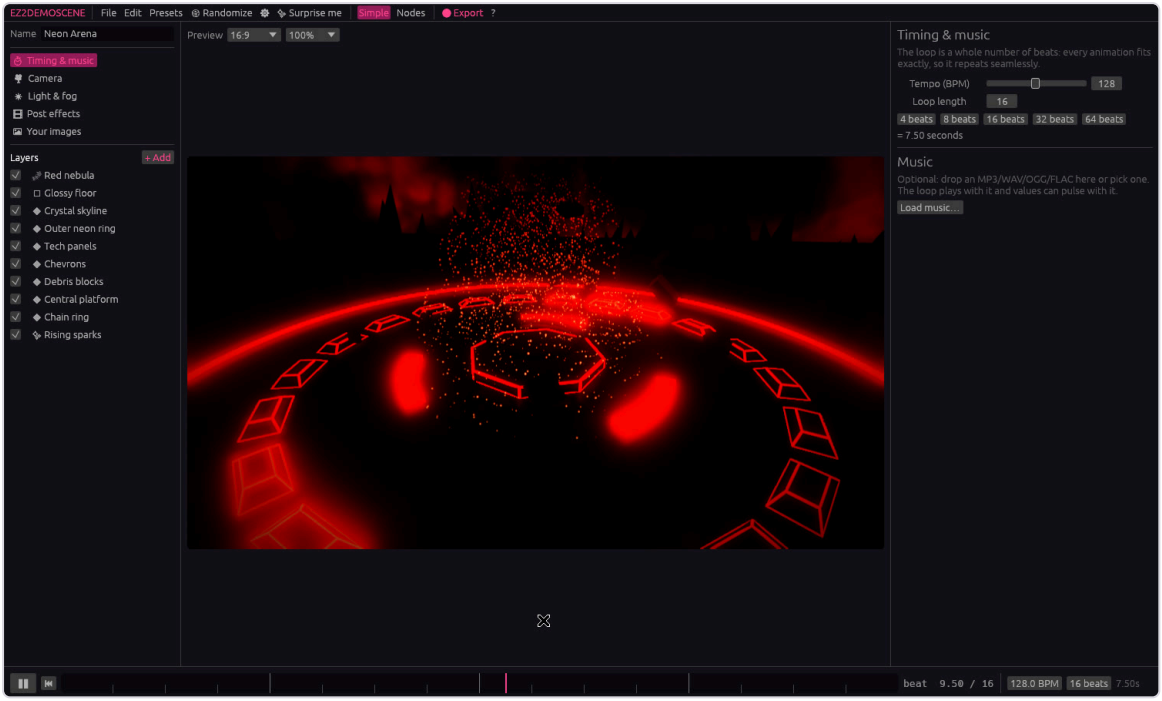
None of these animate by themselves, so loops stay seamless: the wobble and warp depend only on where things are, and a moving camera or spinning shape produces the same wobble at the end of the loop as at the start. The **PSX Crypt** preset shows the PlayStation look (warped brick walls and wooden floor, a candle-lit idol, walls popping open near the camera); **Fog Island** is the Nintendo 64 (soft textures, close fog, video blur) and **Saturn Ghosts** the Saturn (checkerboard see-through ghosts and wisps, pillars vanishing as the camera brushes them). **Slipgate Courtyard** is Quake

(palette lighting, torches and a flickering sky light by light styles, a wobbling lava pool, square embers, the two-layer sky) and **Slime Falls** a turbulent slime canyon and waterfall under a strobing light.

Palettes

Palette	Colours
EGA (16)	
CGA (4)	
Commodore 64 (16)	
Game Boy (4)	
PICO-8 (16)	
Amiga copper (16)	
ZX Spectrum (8)	
VGA cube (216)	Every colour quantised to 6 levels per channel, a subtle retro banding.
Green phosphor (4)	
NES (55)	The Nintendo Entertainment System's fixed palette (the common NTSC approximation): muted blues, greens, browns and pastels.
1-bit black & white (2)	 Macintosh, ZX81; lovely with full dither.
Virtual Boy red (4)	
Amstrad CPC (27), Master System (64), Mega Drive (512), Amiga 12-bit (4096)	Colour cubes: every mix of 3, 4, 8 or 16 levels of red, green and blue, which is what these machines could show. The more levels, the subtler the banding.

10 Timing & music



Timing & music.

Tempo and loop length

Tempo (BPM) and **Loop length** (in beats) set how long the loop is. For example, 128 BPM × 16 beats = 7.5 seconds. The quick buttons set 4, 8, 16, 32 or 64 beats. These values also appear in the timeline.

Because every animation is counted per loop, **changing the BPM speeds the whole scene up or slows it down**, and it stays in sync.

Adding music

Use **Load music…** or drop an MP3, WAV, OGG, FLAC, M4A or AAC file onto the window. The song is **analysed** once when it loads: loudness, kick, bass, mids, highs and brightness curves; the **hits** (kicks, snares and claps, hi-hats, any onset, new notes); a 16-band spectrum; the melody's note; and the **tempo**. The timeline shows the loudness, with kicks as red ticks and snares as yellow ones. The **Meters** show every source live. The analysis runs in the background (**Analysing music… 42%** in the viewport bar), so you can keep editing; music-driven settings start reacting when it finishes, and exports wait for it. Changing only the MIDI file or its offset reuses the analysis instead of starting again.

Loop a part of the song, or the whole song

- **Loop a part of the song** (default): the loop reacts to a window of the song as long as the loop, starting at **Starts at**. The window is treated as a circle: the last moments fade into its start, and a hit near the end keeps ringing after the loop point, so the loop stays seamless. Video exports contain exactly that window of the song, repeated. **Use song tempo** sets the BPM to the

detected tempo and snaps the window onto a bar; **Snap to bar** only snaps; **Fit whole song** makes the whole track one loop.

- **Whole song**: the preview and exports run from the start to the end of the song (repeats don't apply); loop animations keep cycling underneath.

Making values react: the 🎵 row

Every ~ panel has a 🎵 row under the rhythm settings. Pick a source and an amount:

Source	What happens
Follow: Loudness, Kick, Bass, Mids, Highs, Brightness	The value rises with that part of the sound. Tick **smooth** for a calmer curve and use **ignore below** to react only to the loud parts.
Follow: Melody pitch	The melody's note from C (0) to B (11/12). On **Hue shift** with amount 1, each note gets its own colour.
Each kick / snare / hi-hat / hit / new note	Plays a **shape** (a beat fade, e.g. Exp fade out) of the chosen **length in beats** on every hit, scaled by how hard it hit. Flashes on the song's real kicks rather than on the metronome.

A negative amount pulls the value down instead (e.g. the camera distance on each kick, for a punch-in). The curve preview shows the result with the music.

Time warp

Time warp makes motion (spins, orbits, camera moves, scrolling, clouds, LFOs) run faster while the chosen source is strong, so the scene surges on every kick or on the loud parts. The warped time is rescaled so a full loop still ends exactly where it started, so the loop stays seamless. Beat fades, strobes, blinks and shakes stay on the real beat. A negative amount slows motion down on hits.

Equalizer copies

Shapes with copies have an **Equalizer** setting (under Copies → variation): each copy grows upwards and glows with one frequency band, low notes first. 16 copies in a row or an arc make a classic spectrum display (see the **Music Reactor** preset).

MIDI notes

Load MIDI notes… (or drop a .mid file) takes the hits and the melody from a MIDI file instead of guessing them from the audio: drums on channel 10 give exact kicks, snares and hats, other channels give new notes and the melody pitch, with their velocities. **MIDI offset** lines the notes up with the audio. A MIDI file works without an audio file too.

Live input

Live input (microphone / line-in) makes the preview react to what the microphone hears, for VJ sets. Every source works live, and so does time warp. Live input is not repeatable, so exports always use the music file.

> **Tip for musicians**
>
> Load your track, press **Use song tempo**, pick a strong section with **Starts at**, then link the glow of your shapes to **Each kick**. Export with the music included and the visuals land exactly on your beats.

Scenes & timeline

One project can hold several **scenes**, each with its own layers, camera, light and post effects, and play them one after another on a **timeline**. Open 🎬 **Scenes & timeline** on the left and press **Start a timeline**: your scene becomes the first clip.

- **Scenes:** **+ Empty scene** or **+ Copy of this scene**, then ✏ **edit** to work on one; every panel (layers, camera, light, effects, nodes) now edits that scene. Each scene has its **Own loop**, which keeps looping inside its clips.
- **Clips:** pick the scene and the length in beats, reorder with ⬆ ⬇. Each clip comes in with a transition over its first beats: **Cut**, **Crossfade**, **Wipe** (with an angle), **Iris**, **Flash** (to white and back), **Glitch** (blocks of the new scene pop in) or **Cut on kick** (with a song: the cut lands on the first kick).
- The whole timeline is the loop (the Timing panel shows its length), so exports play every clip and loop seamlessly: the first clip's transition comes in from the last one.
- Above the preview, **This scene** shows only the scene you are editing and 🎬 **Timeline** plays everything. The move/rotate handles only show for the scene being edited.
- With a song loaded, 🎵 **Clips from the song's sections** finds where the music changes (drops, breakdowns, new parts) and makes one clip per part, taking turns through your scenes; the loop becomes the whole song from its start.
- **Stop the timeline** goes back to one looping scene; the other scenes are kept.

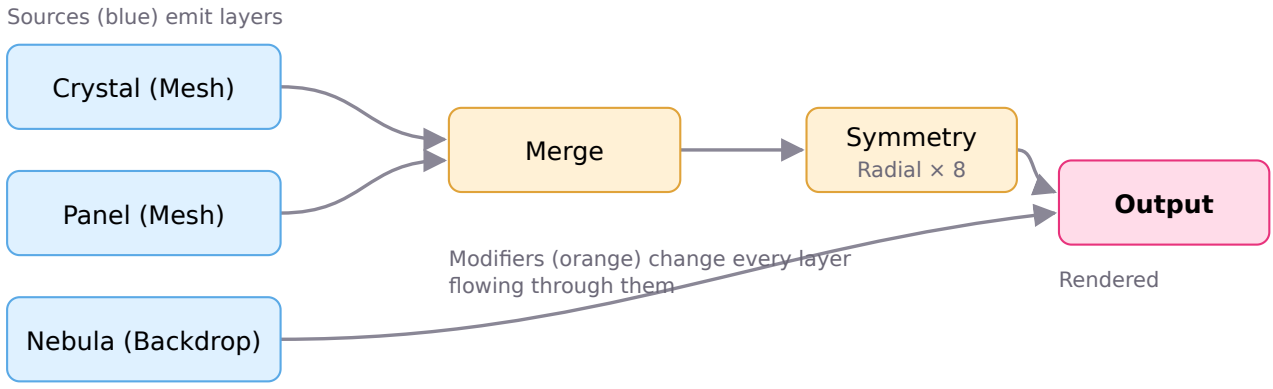
The **Scene Tour** preset chains three scenes with glitch, wipe and iris transitions.

11 Randomize, Surprise me & undo

- 🎲 **Randomize** shuffles the current scene: one harmonious hue rotation of all colours (with a colour scheme on, a new key colour instead), new shapes and layouts, new motion and a few post-effect changes. Every click gives a new variation.
- ⚙ opens the **Randomizer** settings. Choose what may change (Colours, Shapes & layouts, Motion, Post effects), how strongly, and the **seed**. The same seed always gives the same result.
- **Surprise me** picks a random preset and randomizes it strongly.
- **Undo / redo** (`Ctrl`+`Z` / `Ctrl`+`Shift`+`Z`) covers everything, so randomize freely.

12 Nodes mode (advanced)

Click **Nodes** in the top bar to build the scene as a **graph**. Layers become **source nodes** that flow through **modifiers** into an **Output**. Everything connected to Output is rendered.

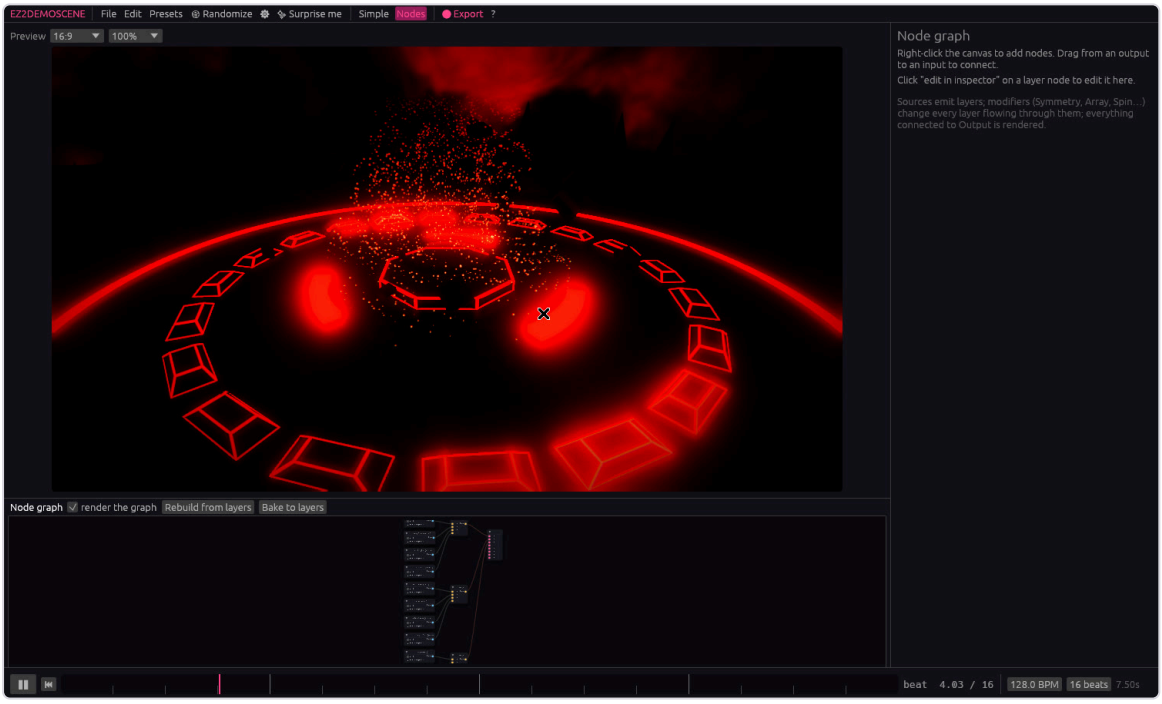


The crystal and panel layers are merged, turned into 8 radial copies, and sent to Output together with the nebula.

Node	What it does
Source	One layer. Click **edit in inspector** to edit it on the right with the usual controls.
Symmetry	Replaces the symmetry of every incoming layer.
Offset	Moves every incoming layer.
Spin	Adds turns per loop.
Scale	Multiplies the size.
Tint	Rotates hues and scales the glow.
Array	Repeats the incoming layers N times, stepping the position and rotation for each copy.
Jitter	Moves and turns every incoming layer by a random amount (**seed** reshuffles). With **re-roll** at 0 it scatters the layers once; above 0 it becomes a beat-synced **shake** that picks a new direction that many times per loop. **Move** and **Turn** have ~, so fades work here too.
Mirror	Adds a mirrored copy of every incoming layer, across the X, Y or Z plane at a chosen position.
Strobe	Sets the Blink / strobe of every incoming layer (chapter 7).
Colour / material	Sets the glow colour, multiplies the glow, and for shapes can set a texture, a neon wireframe look and a glitch.
Deform	Sets the Deform of every incoming shape.
Colours across copies	Sets the colours across the copies of every incoming shape.

Along a curve	Lays the copies along a curve. Wire a **ribbon** layer into its second input (**ribbon**) and they follow that ribbon exactly, wherever it is.
On a surface	Scatters the copies over the shape layer wired into **surface**, following its placement.
On a terrain	Stands the copies on the terrain wired into **terrain**.
Drive	Sets one setting of every incoming layer from a signal (see *Signals* below).
Merge	Combines up to 4 streams.
Output	Up to 8 inputs. Everything here is drawn.

Working with the graph: right-click empty space to add nodes, drag from an output dot to an input dot to connect, and right-click a node to **Duplicate** or **Delete** it. The toolbar has three controls: **render the graph** switches between the graph and the plain layer list; **Rebuild from layers** starts a new graph from your layers; and **Bake to layers** turns the graph's result into ordinary layers and returns to Simple mode.



Nodes mode: the viewport on top, the graph below, the inspector on the right.

Signals: wire any setting to a wave, a beat or the music

Besides layers, wires can carry **signals**: one number that changes over the loop. Signal pins are **green** (outputs are squares). A signal node shows a small graph of its value over the whole loop, with the current moment marked in pink. Add them from the right-click menu under **Signal**.

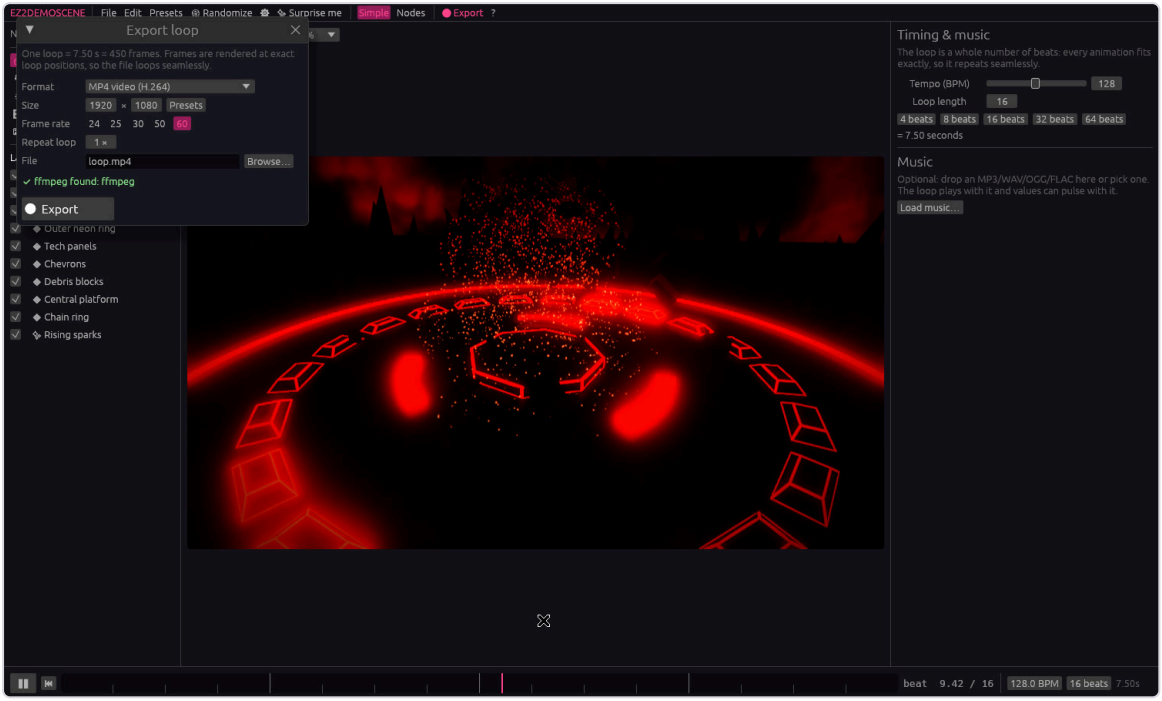
Signal node	What it does
Wave / music	Any animation: a wave with whole cycles per loop, a fade on every beat, random steps, or a link to the music (follow the kick, bass or melody, or react to hits). It uses the same ~ and 🎵 controls as every other value.

Math	A + B, A − B, A × B, A ÷ B, the smaller or larger one, or A > B / A < B (1 or 0). Unconnected inputs use the numbers typed in the node.
Remap	Maps one range to another, e.g. −1…1 to 0.7…1.35. **stay in range** clamps.
Quantize	Rounds to steps (4 steps per 1 → 0, 0.25, 0.5…).
Smooth	Averages over a few beats around each moment. Great on random steps. Music links have their own smoothing (the 🎵 row), which works better for them.
Mix	Blends A to B by T.
Sequence	Steps through your list of values, one every N beats, starting over with the loop. **glide** slides from each value to the next.
Hit counter	Counts kicks (or snares, hats, any hit, notes) since the start of the loop: 0, 1, 2… up to **count to** − 1, then starts again. Through a Remap (0…3 → 0…1) it steps a setting on every kick.

A **Drive** node puts a signal to work. Wire layers into its first input and a signal into the second, then pick the **setting** from the list: `kind.…` are the layer's own settings (for a shape, `kind.material.emissive` is its glow), `transform.…` its placement, and `.0 .1 .2` are x, y, z (or red, green, blue). Choose how the signal is used: **set to** replaces the value, **add** adds to it (the setting keeps its own animation), and **multiply by** scales it. Chain Drive nodes to drive several settings.

Signals are worked out from the loop position and the music at that moment, so graphs always loop seamlessly. The **Signal Flow** preset shows a stepped glow Sequence and a smoothed random size on the centrepiece. **Bake to layers** drops the Drive nodes (settings keep their own values).

13 Exporting your loop



The export dialog (`Ctrl`+`E`).

Option	Notes
Format	**MP4 (H.264)** works everywhere. **WebM (VP9)** is for the web. **Animated GIF** is for chats and forums. **PNG sequence** writes one PNG per frame into a folder and needs no ffmpeg; it is ideal for video editors and VJ software.
Size	Type a size or use **Presets**: 720p, 1080p, 1440p, 4K, Square, Vertical (phone), GIF small, Retro 4:3, and small sizes **for the web**: 640×360, 320×180, square 360×360 and 240×240, vertical 360×640. **Half size** and **Quarter size** shrink whatever size is set and keep its shape.
Frame rate	24, 25, 30, 50 or 60 fps; for GIFs 10, 12.5, 15, 20, 25 or 30 fps. The dialog shows how many frames one loop needs.
Colours *(GIF)*	16, 32, 64, 128 or 256 colours per frame. Fewer colours make a smaller file and a more retro look.
Dither *(GIF)*	On, gradients are smoothed with a fine pattern of dots. Off, they become flat bands of colour, and the file is smaller.
Motion blur	Averages 4, 8 or 16 in-between moments per frame for smooth, film-like motion (fast spins and flying debris smear instead of stepping). **Shutter** sets how much of the time between frames is blurred. The export takes that many times longer, and the video still loops perfectly.
Repeat loop	Makes the file longer by repeating the loop. Useful for platforms that don't loop videos.
Music	Include the music track (MP4/WebM only).

File / Folder	Where to save. Use **Browse…**.

Press ● **Export**. A progress bar shows the frames being rendered, and you can keep working or press **Cancel**. Export renders every frame at full quality (anti-aliasing and full-resolution effects) even if your preview is set lower.

> **GIF tips**
>
> GIFs get big quickly. For a web page or a chat, try a web size such as 320×180 at 15 fps with 64 colours and **Dither** off: that makes a GIF about six times smaller than 480×270 at 25 fps. The smaller size alone halves the file, and 15 fps and 64 colours each take off about a third; dither off saves most with few colours. Keep loops short (8 beats), and avoid film grain, which makes GIFs much larger.

> **"ffmpeg not found"?**
>
> Install ffmpeg, then either make sure `ffmpeg` runs in a terminal or paste the path to the `ffmpeg` program into the box in the export dialog and press **Check**.

Stills

File → Save still image… saves the current frame as a PNG at the export size.

Command line

You can render without opening the window, which is handy for batch jobs:

```
ez2demoscene --list-presets
ez2demoscene --export friend_scene.ez2pack loop.mp4   # packs work too
ez2demoscene --render "Neon Arena" still.png --phase 0.25 --size 1920x1080
ez2demoscene --export "Gold Kaleido Room" loop.mp4 --size 1920x1080 --fps 60 --repeats 4
ez2demoscene --export "Tesla Swarm" smooth.mp4 --fps 30 --motion-blur 8 --shutter 0.5
ez2demoscene --export my.ez2.json frames/          # PNG sequence into a folder
ez2demoscene --export "Neon Arena" web.gif --size 320x180 --fps 15 --colors 64 --no-dither
# small GIF
ez2demoscene --write-presets my_presets/           # presets as editable project files
ez2demoscene --write-textures my_textures/         # the retro texture pack as PNGs
```

`--phase` is a position in the loop from 0 to 1 (0.5 is the middle).

14 Saving, sharing & your library

Project files

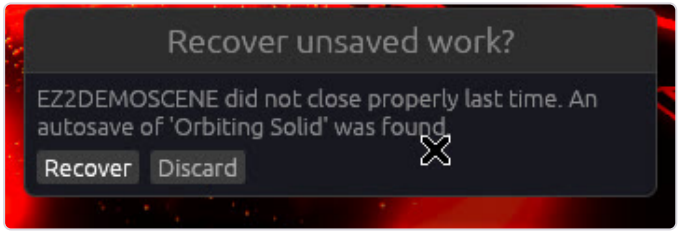
File → Save (`Ctrl`+`S`) writes a `.ez2.json` project. Models, images and music stored **inside the project's folder** (or its sub-folders) are remembered **relative to the project**. You can move or copy the whole folder, even to another computer, and everything is still found. Files elsewhere on your disk are remembered by their full path.

Packs: one file with everything

File → Save as pack (.ez2pack)… writes a single file containing the project **and copies of every model, image and music file it uses**. It's the best way to send a scene to a friend or archive it. Open a pack with **File → Open…** or by dropping it on the window. It is unpacked into your library folder, and **Save** then asks where to keep the project. If some files were missing when packing, the status bar lists them.

Autosave & crash recovery

While you work, unsaved changes are autosaved every 30 seconds. If the program or computer crashes, the next start offers to **recover** them:

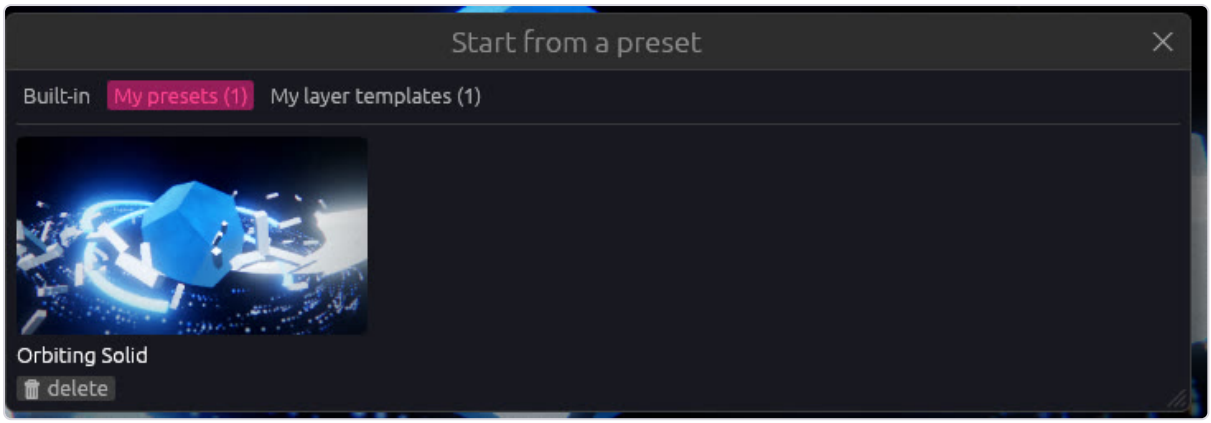


Shown after a crash. **Recover** brings back the scene; **Discard** throws the autosave away.

After recovering, save the project somewhere with `Ctrl`+`S`. When you quit normally, the autosave is deleted.

My presets

File → Save as my preset… adds the current scene to the gallery with a thumbnail of the current frame. Find it under **Presets → My presets**. Clicking one opens it as a new untitled project, so your preset is never overwritten by accident. The 🗑 **delete** button removes it.



The gallery's **My presets** tab.

Layer templates

Built a great particle effect or neon ring? **Right-click the layer → Save as template**. It then appears in **+ Add → ⭐ My templates** in every project, and also in the node graph's right-click menu. Manage your templates in **Presets → My layer templates**.

> **Where is my library?**
>
> Your presets, templates, unpacked packs and autosave live in your user data folder: `%APPDATA%\ez2demoscene\EZ2DEMOSCENE\data` on Windows, `~/.local/share/ez2demoscene` on Linux, and `~/Library/Application Support/org.ez2demoscene.EZ2DEMOSCENE` on macOS. Copy it to back everything up.

15 Phones, tablets & the web

EZ2DEMOSCENE also runs **in a web browser**, with nothing to install. It's the same editor with the same presets, layers, node graph and exports. Open `https://roganis.github.io/ez2demoscene/` in Chrome, Edge or Firefox on a computer, or in Chrome on an Android phone or tablet. On Android you can also install the **EZ2DEMOSCENE app (APK)**, which bundles the same editor and works offline.

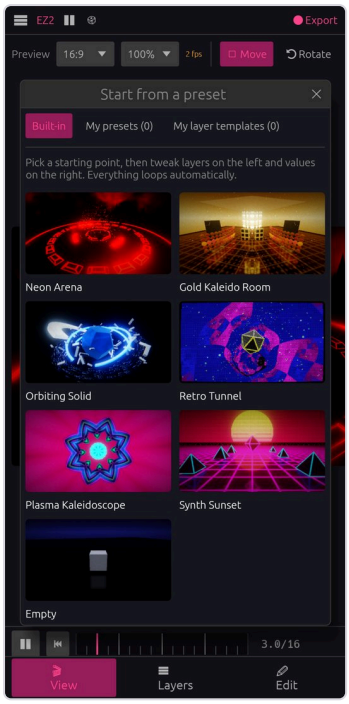
Installing

How	Steps
Browser	Just open the link. The first visit downloads about 5 MB; after that it loads from the cache and also works offline.
Home-screen app (PWA)	In Chrome, open the ⋮ menu → **Add to Home screen** / **Install app**. It opens full-screen like a normal app.
Android APK	Download `EZ2DEMOSCENE-android.apk` (or `EZ2DEMOSCENE-android-signed.apk`, when there is one) from the project's GitHub Releases page, open it and allow "install unknown apps" for your browser or file manager when Android asks.

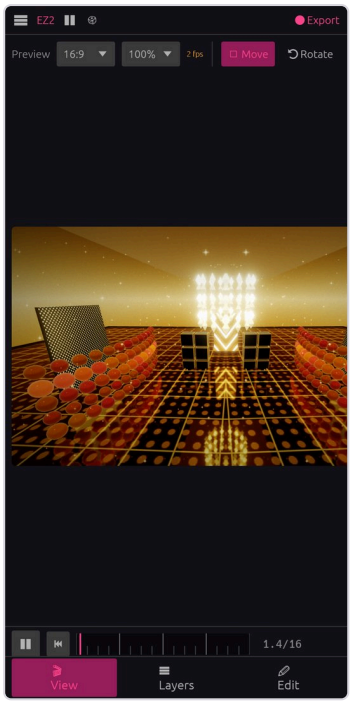
The phone layout

When the window is narrower than about 820 pixels, as on a phone held upright, the editor switches to a **one-panel layout**. The top bar holds ☰ (all menus), play/pause, 🎲 (Surprise me) and **Export**. A compact timeline sits above a **tab bar** with three tabs:

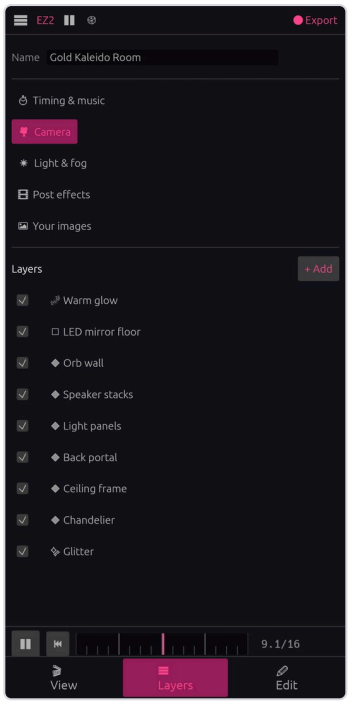
- **View**: the live preview, with the Move/Rotate gizmos.
- **Layers**: the scene settings and layer list (or the **Graph** in Nodes mode).
- **Edit**: the inspector for whatever you selected. Tapping a layer, a scene item or a node jumps here automatically.



Preset gallery on a phone.



The **View** tab.



The **Layers** tab.

Turn a tablet sideways (or use a desktop browser) to get the full desktop layout with all panels side by side.

Touch gestures in the preview

Gesture	What it does
One finger on a gizmo handle	Moves or rotates the selected layer, as with the mouse. Handles are bigger on touch screens.
One finger elsewhere	Orbits the camera around the scene.
Two-finger pinch	Moves the camera closer or further away.
Two-finger twist	Turns the camera around the scene.
Two fingers up/down	Raises or lowers the camera.
Long press on a layer	Opens its menu (duplicate, delete, save as template…), like a right-click.

Files in the browser

- **Importing** models, images and music opens your device's file picker; you can also drop files onto the page on a computer. Imported files are kept inside the browser, so they're still there next time.
- **Saving**: **File → Save** downloads a **pack** (`.ez2pack`) with the scene and all its files. Open it again with **File → Open…**, on the web or in the desktop app. In the Android app, saving and exporting open Android's **Save as** dialog: pick a folder (Downloads, Documents, Drive…) and tap **Save**. From there you can share the file to a chat with the Files app.
- **My presets, layer templates and the autosave** live in the browser's storage (IndexedDB). If you come back with unsaved changes, the editor offers to **recover** them. Clearing the browser's

site data deletes this library, so save packs of anything you want to keep.

Exporting on the web

Format	Notes
MP4 / WebM video	Encoded by the browser's own video encoder (WebCodecs, hardware-accelerated on most phones). Loops stay frame-exact. MP4 uses H.264 where available, otherwise VP9 or AV1. **The web version exports video without the music track**; use the desktop app for videos with sound.
GIF	Works everywhere; loops forever by itself. Keep it small for sharing: pick a size under **Presets → For the web**, a lower frame rate and fewer colours.
PNG frames (.zip)	Every frame as a lossless PNG, for editing elsewhere.

The preview pauses while exporting so the device can concentrate on the frames. Phones are slower than desktop PCs, so start with 720p or less, and keep the tab open until the download starts.

> **WebGPU and WebGL**
>
> On computers, recent Chrome and Edge render with **WebGPU**, the fastest path. Other browsers, and all Android phones and tablets, use **WebGL2**, which looks the same (some phone GPUs corrupt the picture with WebGPU). If a scene stutters, lower **Preview** to 50% in the View tab; exports are unaffected.

Graphics settings

☰ → 🖥 **Graphics** (the 🖥 button on a computer) shows which backend, GPU and driver the app is using; 📋 **Copy** copies the details for a bug report. On the web and in the Android app you can also force **WebGL2** or **WebGPU** there. The change applies after ⟳ **Restart now**. If the chosen backend can't start, the app goes back to **Automatic** by itself.

16 Recipes

Neon arena (black gloss and red neon)

1. Background **Nebula clouds**, with very dark colours A/B and a red C. Set the fog colour to almost-black red.
2. Add a **Mirror floor**: colour black, Reflection 0.35, Blur 0.3.
3. Add **Crystal cluster** with Copies → Radial (count 50, radius 34), Size 7, Random tilt 18, Random size 0.5. Use a glossy black material (metallic 0.9, roughness 0.15, Faceted).
4. Add **Panel** with Radial (20, radius 11) and Stretch 2.6 / 0.35 / 1.2. Glow red, **Edges**, strength 5, and Light chase 1.
5. Add **Ring band** (Width 0.02) stretched 16 / 1 / 16 with red glow, Full, and a Sine animation.
6. Add **Vortex** particles in orange → red. Turn on Bloom 1.1 and Chromatic aberration.

Golden mirrored room

1. Background **Gradient sky** in warm golds. Set the light to warm white and the ambient to 0.45.
2. Add a **Mirror floor** with Reflection 0.85, Reflection tint gold, texture `led_grid` and Grid glow 0.6.
3. Add **Sphere** with Copies → Wall (14 × 5, curve 50°), Size 0.33, Rotation y 90, Position x 6.5, Symmetry **Mirror X**, and Light chase 0.6.
4. Add **Cube** with texture `speaker`, a Grid of 2 × 3 and Symmetry Mirror X. Pulse the glow on the beat.
5. Add **Octahedron** with Spiral copies high above the floor as a chandelier, plus **Snow / glitter** star particles.
6. Post: **Mirror split** left → right and Bloom 1.0.

Crystal orbited by debris

1. Background **Starfield**.
2. Add **Dodecahedron**, Size 1.6, blue, metallic 0.5, roughness 0.18, Faceted, Rim light 0.9, Spin / loop x 1, y 1.
3. Add **Cube** with Copies → Orbit swarm (70, radius 3.2, spread 0.9), Stretch 0.12 / 0.12 / 0.5 and Random size 0.6.
4. Add **Ring band** arcs (Arc 250°) with blue glow and Spin / loop y 2, plus **Ring orbit** particles with trails.

90s demo tunnel

1. Background **Raymarched tunnel** with Wall texture `xor` and Motion / loop 4.
2. Add **Icosahedron** with texture `checker`, Chunky pixels, Glow Edges, and Spin / loop 2 / 1 / 0.
3. Add **Warp stars** with the Pixel square sprite and Trail 4.
4. Post: Pixelate 3, Retro palette EGA with dithering 0.7, CRT monitor, and Beat flash 0.15.

Stormy lake

1. Background **Volumetric clouds**, variant **Storm**, with grey-blue colours. Fog colour dark grey-blue, Sun strength 0.7.
2. Add **Terrain**: Solid, Shape **Ridged mountains**, Biome **Alpine**, Valley 0.45, and Liquid **Water** at Level 0.12.
3. Add **Weather → Rain** with Falls / loop 14 and a Sine on Wind for gusts. Tick **Lightning**.
4. Post: lower Saturation to 0.8 and add a little Film grain.

Lava world

1. Background **Volumetric clouds** (Overcast) in dark reds, with the sun low in front of the camera.
2. Add **Terrain**: Shape **Canyons**, Biome **Volcanic**, Liquid **Lava** at Level 0.3, Current / loop 2. Give the lava Glow a slow Sine.
3. Add **Weather → Rising embers** with a strong Brightness so they bloom.
4. Post: **God rays** with a warm tint, and Bloom.

Sunken temple (underwater)

1. Background **Gradient sky** from light blue (top) to deep blue. Fog colour deep blue, Fog density 0.04.
2. Turn on **Underwater caustics** (Amount 1.6, pale blue).
3. Add a **Mirror floor** with Reflection 0.05 and a sandy colour, pillars (Cylinder, Radial copies) and a glowing gem.
4. Add **Fountain** particles with the Ring sprite as bubbles, and **God rays** with a low threshold.

17 Reference

Keyboard & mouse

Action	Shortcut
Play / pause	`Space`
Save	`Ctrl` + `S`
Open	`Ctrl` + `O`
Export	`Ctrl` + `E`
Undo / Redo	`Ctrl` + `Z` / `Ctrl` + `Shift` + `Z` (or `Ctrl` + `Y`)
Move / Rotate / Scale tool	`W` / `E` / `R`
Ground grid on/off	`G`
Select an object	Click it in the viewport
Snap while dragging a handle	Hold `Ctrl`
Turn the camera	Drag in the viewport (not on a handle)
Zoom the camera	Mouse wheel over the viewport
Scrub time	Click or drag in the timeline
Import	Drop project, model, image or music files onto the window

Performance

The small **fps** number above the viewport shows how smoothly the preview runs. It turns orange below 24 fps. **Hover it** to see the frame time, triangle and particle counts, whether the mirror floor is drawing the scene a second time, and the **five most expensive layers**. A ⚠ next to a layer in the list marks a heavy layer. Hover it for details. The usual fixes are fewer copies, less sphere detail, fewer particles or a shorter trail. You can also lower the preview quality, which only affects the preview and never your exports. Volumetric clouds and the raymarched backgrounds are the heaviest layers; their **Quality** setting trades detail for speed.

Troubleshooting

Problem	Solution
Playback stutters	Hover the fps readout to find the heaviest layers (see Performance above), or lower the preview quality. The fractal background and big particle counts are the heaviest. Exports are not affected.
The computer crashed and I lost my work	Start EZ2DEMOSCENE again and choose **Recover** (chapter 14).

A friend's project can't find its files	Ask them to send a **pack** (**File → Save as pack**) instead of the `.ez2.json` file.
Everything is washed out or one colour	A big glowing object is probably filling the screen. Reduce Glow, raise the Bloom threshold, or check the Size/Stretch of ring and panel layers.
A shape is invisible	Check its visibility box, its Size, whether it is below the floor or inside another object, and the fog density.
My model shows as a cube	The file couldn't be read. The reason appears in red at the bottom of the inspector. Use **Edit → Reload models & images** after fixing the file.
No sound	No audio output device was found. The music is still used for sync, ♪ modulation and export.
Video export fails	Check ffmpeg (chapter 13), or export a PNG sequence instead.
The picture corrupts or flickers on a phone or in a browser	Open 🖥 **Graphics** and switch to the other backend (WebGL2 or WebGPU), then restart (chapter 15).
The app doesn't start	Update your graphics drivers. On Linux, install Vulkan drivers (e.g. `mesa-vulkan-drivers`).

Project files

Projects are saved as readable `.ez2.json` text files. Imported models, images and music are stored as **file paths**, so keep the files inside the project's folder (they're then stored relative to it), or share a **pack** (chapter 14). The built-in presets are also available as project files in `assets/presets`.

Glossary

Term	Meaning
Beat / BPM	A musical pulse. BPM is beats per minute.
Loop / phase	The repeating clip. Phase is a position in it, from 0 (start) to 1 (end, which equals the start).
Instancing	Drawing many copies of one shape efficiently.
Emission / glow	Light a surface gives off by itself.
Bloom	The halo that bright things spill onto their surroundings.
Raymarching	A technique that draws shapes defined by maths (tunnels, fractals) instead of polygons. It is a demoscene classic.
Dithering	Patterns of pixels that fake more colours than a palette has.
Demoscene	A computer art subculture, dating back to the 1980s, that makes real-time audiovisual "demos".

EZ2DEMOSCENE is free software under the GPL-3.0-or-later licence. The built-in texture pack is CC0.